

Two-Sided Markets and Restricted Boltzmann Machines

Tetsuya Hoshino & Romans Pancs*

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Abstract

We extend the standard model of two-sided markets (e.g., Uber or Upwork) by allowing benefits from pairwise interactions to differ across pairs of agents. We show how to estimate these benefits consistently and tractably by observing agents' repeated decisions whether to participate in a market. The proposed estimation technique mitigates the combinatorial explosion due to heterogeneous interaction effects by exploiting a formal connection to a prominent class of neural networks called restricted Boltzmann machines. We illustrate the value of learning model parameters by exhibiting a simple pricing scheme that, in a special case, maximizes both welfare and revenue.

Keywords: two-sided markets, platforms, restricted Boltzmann machines, contrastive divergence

JEL Classification Numbers: C45, D85, L10

1 Introduction

Two-sided markets balance supply and demand across two disjoint groups of participants, or sides. Ride-sharing platforms Uber and Lyft, freelancing platforms Upwork and Fiverr, and online-dating platforms Match.com and Tinder balance riders and drivers,

*Hoshino (tetsuyahoshino137@gmail.com) is at ITAM. Pancs (rpancs@gmail.com) is at the University of Rochester and ITAM. For feedback, we thank Nageeb Ali, Lester Chan, Ben Golub, Igal Hendel, Glen Weyl, Soomin Jung, Asen Kochov, Shaun McRae, Alessandro Pavan, and Yuta Watabe, as well as the workshop and seminar audiences at Northwestern, Penn State, the University of Toronto, the 2023 SAET Conference, and the 2024 ESIF Economics and AI+ML Meeting.

clients and freelancers, and men and women, respectively. The literature on two-sided markets, pioneered by [Caillaud and Jullien \(2003\)](#) and [Rochet and Tirole \(2003\)](#) and surveyed by [Jullien, Pavan and Rysman \(2021\)](#), examines platform pricing and regulation. This literature maintains the assumption that network effects are homogeneous: each agent cares only about the number of participants on the other side but not about these participants' identities. While this assumption may be valid for ride-sharing platforms, it is likely inadequate for freelancing platforms, in which disparate skills, expertise, and time commitments matter greatly, and for online dating platforms, in which different users care about different characteristics of partners. Our contribution is to extend the standard two-sided markets model to accommodate heterogeneous network effects. (Heterogeneous network effects in other contexts have been examined by [Gomez and Pavan, 2016](#).)

We organize our analysis around the problem of a startup that launches a novel two-sided platform. First, the startup must learn the demand for the platform's services by learning the model's parameters, which is done by observing agents' repeated decisions about whether to log into the platform. To motivate the value from this learning, we assume that the startup must then compute optimal prices, to maximize welfare or revenue. Because network effects are assumed to be heterogeneous, both tasks present tractability challenges, even from the numerical perspective, as we show. This paper's main contribution is in tackling the first task by proposing a numerically tractable estimator of the model's parameters. As proof of concept, we tackle the second task in a special case that, in spite of being special, retains the heterogeneity of network effects.

Our model of the two-sided market extends the classic (all-to-all matching) model of [Armstrong \(2006\)](#) by introducing heterogeneous network effects. With prices fixed, our model is a coordination game in which n_1 agents on side 1 and n_2 agents on side 2 of the market each take a binary decision about whether to log into the platform. The profile of these participation decisions is denoted by the vector $\mathbf{x} \equiv (\mathbf{x}_1, \mathbf{x}_2) \in \{0, 1\}^{n_1} \times \{0, 1\}^{n_2}$. If agents i on side 1 and j on side 2 both participate (i.e., if $x_{1i} = x_{2j} = 1$), then each of them enjoys the same bilateral interaction benefit W_{ij} , which can be positive, negative, or zero. The matrix of interaction benefits—or network effects—is denoted by $\mathbf{W} \in \mathbb{R}^{n_1 \times n_2}$. Regardless of others' participation decisions, each agent i on side 1 of the market enjoys a standalone benefit b_{1i} from participation whenever he chooses to participate (i.e., whenever $x_{1i} = 1$). This benefit, too, can be positive, negative, or zero. We define b_{2j} analogously and collect side-1 and side-2 standalone benefits into the vector $\mathbf{b} \equiv (\mathbf{b}_1, \mathbf{b}_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$. The bipartite graph in [Figure 1](#) organizes the model parameters. Matching is all-to-all because all participating agents on one side of the market bene-

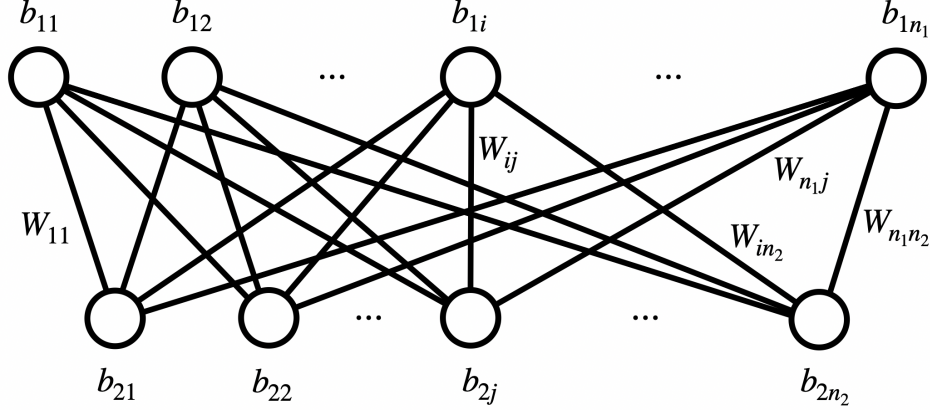


Figure 1: **Payoff parameters in a two-sided market.** The two rows of nodes are the two sides of markets. An edge connects every agent i on side 1 to every agent j on side 2, with the weight W_{ij} (which can be positive, negative, or zero) corresponding to the interaction benefit for each member of the participating pair. There are no interaction benefits for agents on the same side of the market. Agents' standalone benefits from participation, b_{1i} or b_{2j} , label the nodes.

fit from all participating agents on the other side of the market. The all-to-all assumption can be interpreted either literally or as a reduced form for the stochastic many-to-many matching whereby every pair of participating agents are matched with some probability.

Our equilibrium concept is the stochastic best-response equilibrium (sBRE) defined as the limit of agents' "noisy" best-response dynamics in participation decisions (Blume, 1993; Kandori, Mailath and Rob, 1993; Young, 1993; Mele, 2017). In order to myopically best respond to others' current participation decisions, a randomly called-upon agent only needs to know his standalone value from participation and his benefit from interacting with others. Even though market participation is a coordination game, its sBRE is unique as a consequence of the ergodic theorem. At the unique sBRE, two agents' participation decisions are correlated conditional on others' participation decisions if and only if these two agents experience a nonzero benefit from interaction, with the sign of the benefit corresponding to the sign of the correlation. Under appropriate assumptions on the probability distribution of the "noise" (or mistakes) in agents' best responses, the equilibrium probability of a participation profile $\mathbf{x}$ is the logit distribution (Proposition 1):

$$\mathbb{P}(\mathbf{x} \mid \mathbf{W}, \mathbf{b}) = \frac{e^{\mathbf{b}^\top \mathbf{x} + \mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2}}{\sum_{\mathbf{y} \in \{0,1\}^{n_1+n_2}} e^{\mathbf{b}^\top \mathbf{y} + \mathbf{y}_1^\top \mathbf{W} \mathbf{y}_2}}. \quad (1)$$

The startup's first task is to figure out how to learn the model parameters $(\mathbf{W}, \mathbf{b})$ in a tractable manner by observing a sequence of agents' participation profiles, assumed to

be drawn i.i.d. from the sBRE distribution. For example, an ideal data set may come from a dating platform that matches men and women and comprise the online–offline status of every subscriber at fifteen-minute intervals. If the data reveal, say, that college students log in on weekday nights, and young professionals do so on weekend mornings, then the startup may conclude that individuals benefit from matching with similar others. The assumption that the draws from the sBRE are independent is not essential for the consistency of the estimation approach that we advocate (Levine, 1983).

It is tempting to aim for maximum-likelihood estimates for $(\mathbf{W}, \mathbf{b})$ by assembling the probabilities given by (1) into a likelihood function and then maximizing this function by gradient ascent. This approach runs into combinatorial explosion: How should one evaluate (1) when the sum in the denominator has $2^{n_1+n_2}$ terms, a formidable number even for moderate values of $n_1 + n_2$ (e.g., $2^{30} \approx 1,000,000,000$)? And if one cannot evaluate (1) even once, how can one hope to do so over and over again so as to numerically maximize the likelihood?

The second task in the startup’s decision problem, that of optimal pricing, appears to be similarly intractable. Even the problem of evaluating—let alone maximizing—expected welfare or expected revenue runs into combinatorial explosion. For instance, in order to compute the expected welfare

$$\sum_{\mathbf{x} \in \{0,1\}^{n_1+n_2}} (\text{probability of } \mathbf{x} \text{ given prices}) \times (\text{welfare at } \mathbf{x}),$$

one must add up the sum’s $2^{n_1+n_2}$ terms, each of which contains $2^{n_1+n_2}$ terms more in the denominator of the expression for (probability of $\mathbf{x}$ given prices). This expression is derived from (1): for any matrix $\mathbf{P} \in \mathbb{R}^{n_1 \times n_2}$ of pair-specific interaction prices, and for any vector $\mathbf{p} \in \mathbb{R}^{n_1+n_2}$ of individualized participation prices, (probability of $\mathbf{x}$ given prices) = $\mathbb{P}(\mathbf{x} \mid \mathbf{W} - \mathbf{P}, \mathbf{b} - \mathbf{p})$.

Models of two-sided markets in the tradition of Armstrong (2006), with homogeneous network effects, do not suffer from the combinatorial explosion in pricing. What happens if the startup willfully neglects heterogeneity and prices under the misspecified model that assumes homogeneous network effects? Betting on the misspecified model may cost the startup dearly. We show that, under the misspecified model, not only will the startup’s inference about the model parameters be wrong, but also the realized revenue, induced by misguided pricing decisions, may be an arbitrarily small fraction of the optimal revenue (Proposition 5). The startup has no choice but to tackle the model with heterogeneous network effects head-on.

Help arrives from an unexpected place. It turns out that the equilibrium distribution over $\mathbf{x} \equiv (\mathbf{x}_1, \mathbf{x}_2)$ in (1) is known in machine learning by the name “restricted Boltzmann machine (RBM)” (Ackley, Hinton and Sejnowski, 1985; Smolensky, 1986), a special case of the Boltzmann machine pioneered in Geoffrey Hinton’s Nobel prize-winning work. Here, “Boltzmann” is another way of saying logit. “Restricted” means that any two components of a vector $\mathbf{x}_1$ are statistically independent conditional on the values of $\mathbf{x}_2$; *mutatis mutandis* for any two components of $\mathbf{x}_2$. “Machine” is an honorific that indicates that the distribution in (1) has a plethora of uses in machine learning. Indeed, an RBM can recommend movies (Salakhutdinov, Mnih and Hinton, 2007), recognize images (Larochelle and Bengio, 2008), and discover drugs (Wang and Zeng, 2013).

In machine learning, RBMs are trained. To train an RBM means to fit the values of $(\mathbf{W}, \mathbf{b})$ that make the probability distribution in (1) resemble the empirical distribution of the data. Contrastive divergence (CD)—again, proposed by Hinton (2002)—is the leading algorithm for training RBMs. A seemingly unprincipled corruption of likelihood maximization by gradient ascent, CD circumvents combinatorial explosion and is responsible for the resurgence of machine learning in the early 2000s. While a good training algorithm ensures soon enough that the trained distribution fits the empirical distribution closely enough, it is not usually tasked with uncovering underlying parameter values. In fact, the underlying parameter values are of little interest in machine learning because RBM is not a structural model of anything, and its parameters admit no interpretation. Not so in economics.

The startup’s learning problem is more demanding than merely fitting a distribution. One must know the underlying parameters $(\mathbf{W}, \mathbf{b})$ to price well. Therefore, to be of use, CD must estimate $(\mathbf{W}, \mathbf{b})$ consistently. CD’s consistency has received attention from statisticians, on whose results we rely. We show that Jiang, Wu, Jin and Wong’s (2018) theorem for the consistency of CD applies in the context of our two-sided market model (Proposition 2). The model’s special feature that makes Jiang, Wu, Jin and Wong’s results applicable is the (natural) assumption that the startup observes participation decisions on both sides of the market: that is, the entire vector $\mathbf{x}$ is observed. By contrast, most applications of RBMs in machine learning assume that, say, only $\mathbf{x}_1$ is observed, whereas $\mathbf{x}_2$ is hidden.

While its consistency is a theorem, CD’s numerical tractability is an empirical fact supported by over two decades of successful training of neural networks and by the simulations reported later in the paper.

Nothing we know about the theory and practice of RBMs appears to help us tackle the combinatorial explosion in the optimal-pricing problem. Therefore, to illustrate the value

of learning $(\mathbf{W}, \mathbf{b})$, we focus on the pricing problem’s special case in which the noise in agents’ best responses is small. This restrictive case has been the focus of the pricing literature up to now. We construct a class of pricing schemes each of which achieves the first-best (utilitarian) welfare (Proposition 4) and then one that also maximizes revenue by fully extracting the first-best welfare (Proposition 6). Revenue maximization is of interest to the startup. Welfare maximization is of interest to a regulator: without corrective pricing, equilibrium welfare can be an arbitrarily small fraction of the first-best welfare, as we show (Proposition 3).

Our results extend beyond two-sided markets. A three-sided market comprised of, say, dealers who mediate between buyers and sellers is, we argue, a special case of a two-sided market. Our estimation and pricing results continue to apply—as they do to four-sided, five-sided, etc., markets. Another special case of our model is a pair of two-sided markets (e.g., Uber and Lyft) that compete for participants on one of the sides (e.g., riders), while participants on the other side (e.g., drivers) are assumed to be locked into one of the two platforms. Our results carry over once again.

Related Literature

While we engage with the relevant papers as the story develops, below we highlight three broad strands of the literature to which this paper contributes.

Igami (2020) and Samuelson and Steiner (2023) have pioneered the literature that recognizes economic structures in seemingly unrelated machine-learning models. Igami (2020) observes that machine-learning algorithms behind the Deep Blue expert system for chess, the Bonanza engine for shogi (Japanese chess), and the AlphaGo system for Go are related to estimation techniques in dynamic structural econometrics because both the algorithms and the estimation techniques tackle dynamic optimization problems. Samuelson and Steiner (2023) study an economic growth-maximizing policy and relate it to the problem of predictive coding in machine learning. Analogies between problems in asset pricing and machine-learning have been established by Avramov, Cheng and Metzker (2022), Chen, Pelger and Zhu (2024), and Su, Tretyakov and Newton (2024). We likewise discover a novel connection: between an equilibrium outcome in an economic model and activation patterns in an (ontologically unrelated) canonical neural network. This discovery opens up the possibility for repurposing machine-learning results for economic use, and, indeed, we repurpose one, on consistent estimation.

Our paper draws inspiration from Chan’s (2021) incisive observation that a two-sided market with homogeneous network effects is a potential game. We notice that Chan’s potential function coincides with the negative of the “energy function” in a correspond-

ing restricted Boltzmann machine. From here, we conjecture and verify that a model of two-sided markets with heterogeneous network effects, too, admits a potential, and that this potential coincides with the energy function of a restricted Boltzmann machine. Our equilibrium concept, a probability distribution over participation profiles, has [Chan's](#) potential-maximizing Nash equilibrium as the modal outcome. Our equilibrium concept illustrates [Babichenko and Tamuz's](#) (2016, Proposition 4.6) observation that every graphical potential game (here, the two-sided market) ought to correspond to some Markov random field (here, the restricted Boltzmann machine). Our contribution is thus primarily economic-theoretical: we establish and exploit a theoretical interface between an economic model (a two-sided market) and a machine-learning model (RBM) complete with a training algorithm (CD).

[Mele](#) (2017) seminal work is concerned with parameter identification and estimation in a different setting. He identifies model parameters off a large (dense) network when agents engage in stochastic best-response dynamics, and the researcher observes the network just once. His estimation technique of choice is Markov Chain Monte Carlo (MCMC). By contrast, we identify model parameters off repeated observations of the same (bipartite) network and advocate CD. In large RBMs, as an empirical matter, MCMC is prohibitively slow.

Price discrimination in the context of two-sided markets with heterogeneous participants was recently studied by [de Corniere, Mantovani and Shekhar](#) (2025). The importance of heterogeneous users has been documented empirically by [Farronato, Fong and Fradkin](#) (2024).

2 The Model

Conforming with [Armstrong's](#) (2006) workhorse model, our market has two sides, with n_1 agents on side 1, n_2 agents on side 2, and $n \equiv n_1 + n_2$ agents in total. Each agent on side 1 takes the binary decision $x_{1i} \in \{0, 1\}$ whether to join the platform, with $x_{1i} = 1$ corresponding to joining and $x_{1i} = 0$ corresponding to not joining. Let $\mathbf{x}_1 \equiv (x_{1i})_{i=1}^{n_1} \in \{0, 1\}^{n_1}$ denote the vector of participation decisions for side 1, let $\mathbf{x}_2 \equiv (x_{2j})_{j=1}^{n_2} \in \{0, 1\}^{n_2}$ denote the corresponding vector for side 2, and let $\mathbf{x} \equiv (\mathbf{x}_1, \mathbf{x}_2) \in \{0, 1\}^n$ denote the vector that is the entire participation profile.

Agent i on side 1 enjoys the standalone value $b_{1i} \in \mathbb{R}$ from participation, which can be positive, negative, or zero. Let $\mathbf{b}_1 \equiv (b_{1i})_{i=1}^{n_1}$ denote the vector of standalone values for side 1. Define $\mathbf{b}_2 \equiv (b_{2j})_{j=1}^{n_2}$ analogously. If agents i and j on sides 1 and 2, respectively, both participate, then, in addition to the standalone value, each of them enjoys the same

interaction value $W_{ij} \in \mathbb{R}$, which can be positive, negative, or zero. (As we discuss shortly and then in Appendix A.5, the assumption that a pair of agents split the bilateral surplus from interaction equally can be relaxed.) Let $\mathbf{W} \equiv (W_{ij})_{i,j}$ denote the $n_1 \times n_2$ matrix of interaction effects. The vector of interaction effects faced by agent i on side 1 is denoted by $\mathbf{W}_{i\bullet} \equiv (W_{ij})_{j=1}^{n_2}$. The vector of interaction effects faced by agent j on side 2 is denoted by $\mathbf{W}_{\bullet j} \equiv (W_{ij})_{i=1}^{n_1}$.

Fix a participation profile $\mathbf{x}$. The payoffs of an agent i on side 1 and an agent j on side 2 are, respectively,

$$x_{1i} \left(b_{1i} + \mathbf{W}_{i\bullet}^\top \mathbf{x}_2 \right) \quad \text{and} \quad x_{2j} \left(b_{2j} + \mathbf{W}_{\bullet j}^\top \mathbf{x}_1 \right),$$

where the subscript “ $\top$ ” is the transpose operator. From each agent’s perspective, the maximization of his payoff with respect to his participation decision is equivalent to the maximization of the function $\mathbf{b}^\top \mathbf{x} + \mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2$ with respect to the same decision. Games that admit such a function are called potential games, and the common maximand is called the **potential** (Monderer and Shapley, 1996).

Discussion of the Model’s Assumptions

The matching all-to-all protocol implicit in the definition of payoffs is degenerate: everyone on one side of the market matches to everyone on the other side. Or such is the literal interpretation of the model. Equivalently, one can interpret the parameter W_{ij} as the expected benefit from the bilateral interaction that occurs with some probability whenever both i and j participate. This interaction probability can depend on the underlying benefit from interaction, so that, say, the pairs that value interaction more are more likely to match. What is ruled out is the dependence of this probability and of the underlying benefit on who else shows up. Formally, $\mathbf{W}$ does not depend on $\mathbf{x}$.

The assumption that agent i enjoys interacting with agent j exactly as much as j enjoys interacting with i is restrictive. The assumption describes situations in which fairness, legal norms, or equal bargaining powers dictate that the total surplus from the bilateral interaction, denoted by $2W_{ij}$, be split equally, with each agent getting W_{ij} . It is straightforward to extend the model by assuming that all side-1 agents capture, say, 30% of the bilateral surplus, and side-2 agents capture the remaining 70%. The cost of accommodating this extension is only additional notation. A further extension that would allow the benefits from interaction to differ both within and across pairs in arbitrary ways cannot be analyzed using the tools of this paper.

3 Equilibrium

Our equilibrium concept is the **stochastic best-response equilibrium (sBRE)**. It is defined as the limit distribution of stochastic best-response dynamics, in which randomly selected agents myopically update their strategies given other agents' actions (Blume, 1993; Kandori, Mailath and Rob, 1993; Young, 1993; Mele, 2017). The dynamics plays out in discrete steps, indexed by s . The dynamics is assembled from three components: an initial participation profile $\mathbf{x}^{[0]} \in \{0, 1\}^n$, an action revision protocol that specifies which agents are called upon to revise their actions and when, and a stochastic best-response rule according to which the called-upon agent or agents revise their actions.

The first component—the initial participation profile $\mathbf{x}^{[0]}$ —will not affect the limit distribution.

Any revision protocol in a certain class will lead to the same limit distribution. This class of protocols is embedded in the equilibrium definition presented shortly (Definition 1) and includes two prominent members: random scanning and blocked sampling. **Random scanning** selects an agent uniformly at random and asks him to revise his action. Random scanning avoids seemingly arbitrary assumptions on the correlation in agents' revisions and, perhaps crucially, applies to a large class of models. **Blocked sampling** first selects all agents on either side of the market to simultaneously revise their actions, and then selects all agents on the other side to do so. Blocked sampling relies on the market's bipartite structure and captures the idea that, after a revision on side 1, everyone on side 2 wishes to revise before anyone on side 1 would.

At each step $s \in \{1, 2, \dots\}$ of the stochastic **best-response dynamics**, every called-upon agent i on side 1 of the market draws a standard logistic random variable $\varepsilon_{1i}^{[s]}$ that is independent of all past draws and of other agents' contemporaneous draws (if any). He then stochastically best responds to side 2's past actions $\mathbf{x}_2^{[s-1]}$ by revising his binary participation decision $x_{1i}^{[s]}$ according to

$$x_{1i}^{[s]} = \begin{cases} 1 & \text{if } b_{1i} + \mathbf{W}_{i\bullet}^\top \mathbf{x}_2^{[s-1]} + \sigma \varepsilon_{1i}^{[s]} > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (2)$$

where σ is a positive scalar, which, in contrast to Kandori, Mailath and Rob (1993) and Young (1993), we do not insist on sending to zero. Every called-upon agent j on side 2

best responds analogously:

$$x_{2j}^{[s]} = \begin{cases} 1 & \text{if } b_{2j} + \mathbf{W}_{\bullet j}^\top \mathbf{x}_1^{[s-1]} + \sigma \varepsilon_{2j}^{[s]} > 0 \\ 0 & \text{otherwise} \end{cases}.$$

We interpret the stochasticity in best responses as mistakes. The alternative interpretation of the epsilons ($\varepsilon_{1i}^{[s]}$ and $\varepsilon_{2j}^{[s]}$) in (2) as shocks to utility is compatible with our analysis of parameter estimation but is incompatible with our foray into pricing.

We can now assemble the parts.

Definition 1. A **stochastic best-response equilibrium (sBRE)** is the stationary probability distribution of the Markov chain in agents' participation decisions induced by the stochastic best-response dynamics at whose every step,

- an irreducible (but possibly periodic) two-state Markov chain selects a side of the market;
- on the selected side of the market, a subset of agents are chosen according to some fixed (for that side of the market) probability distribution that chooses every agent with a positive probability;
- each chosen agent stochastically best-responds to others' past actions.

The action revision protocol in Definition 1 is the aforementioned random scanning if, at each step, side 1 is selected with the probability n_1/n , and side 2 is selected with the complementary probability n_2/n , and then one of the agents on the selected side is chosen uniformly at random. The protocol is the aforementioned blocked sampling if side 1 is selected at odd steps, and side 2 is selected at even steps, and all agents on the selected side are chosen. Definition 1 accommodates much more general revision protocols than these two, though. It admits persistence and arbitrary imbalance in selecting the side of the market that gets to revise, and it permits different agents to be chosen with different probabilities and in a correlated manner.

Proposition 1 (Equilibrium Characterization). *The stochastic best-response equilibrium exists, is unique, and, to each participation profile $\mathbf{x}$, assigns the probability*

$$\mathbb{P}(\mathbf{x} \mid \mathbf{W}, \mathbf{b}) = \frac{\exp\left(\frac{1}{\sigma} (\mathbf{b}^\top \mathbf{x} + \mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2)\right)}{\sum_{\mathbf{y} \in \{0,1\}^n} \exp\left(\frac{1}{\sigma} (\mathbf{b}^\top \mathbf{y} + \mathbf{y}_1^\top \mathbf{W} \mathbf{y}_2)\right)}. \quad (3)$$

It is known that the logit distribution in (3) is the unique stationary distribution under random scanning and best-response dynamics in potential games, such as ours (see, e.g., [Blume, 1993](#), and [Mele, 2017](#)). Proposition 1 asserts that the same remains true under a great variety of revision protocols. Although we believe a version of Proposition 1 should be known, we have not found a reference to the proof, and, therefore, provide one in Appendix A.1.

Stochastic Best-Response Equilibrium in Context

sBRE’s four critical features will enable us to quickly learn model parameters by observing i.i.d. draws from the equilibrium probability distribution of participation profiles:

- The participation decisions of interacting agents are correlated, and more so for the agents who benefit from interacting more. Correlation reveals the magnitude of network effects.
- The equilibrium distribution has full support; that is, all participation profiles are possible. As a result, the data exhibit rich enough correlation structures to identify all network effects.
- The equilibrium is unique. Uniqueness delivers point (rather than set) identification of model parameters.
- The probability distribution of participation profiles is logit. The logit functional form lends the model tractability, which is crucial for fast estimation.

Each of these features can be motivated by some alternative equilibrium concepts and is inconsistent with some others. We shall now place sBRE in context of these alternatives. The Venn diagram in Figure 2 is a roadmap for the discussion.

The best-response dynamics that undergirds sBRE effectively enables agents to spy on each other’s actions before choosing their own actions. Spying begets correlation. This correlation is stronger when the benefit from the bilateral interaction is larger and, so, is likelier to swamp the random trembles—the epsilons—in best responses. An alternative way to operationalize spying is [Denti’s \(2023\)](#) rational inattention equilibrium (RIE) (which [Denti](#) calls “equilibrium strongly robust to information acquisition”). At an RIE, agents simultaneously acquire noisy signals about others’ equilibrium actions in a fixed-point manner reminiscent of the rational expectations equilibrium with private information à la [Grossman and Stiglitz \(1980\)](#) ([Hebert and La’O, 2023](#)). Compared to sBRE, RIE suffers from two limitations: the multiplicity of RIEs hampers point identifica-

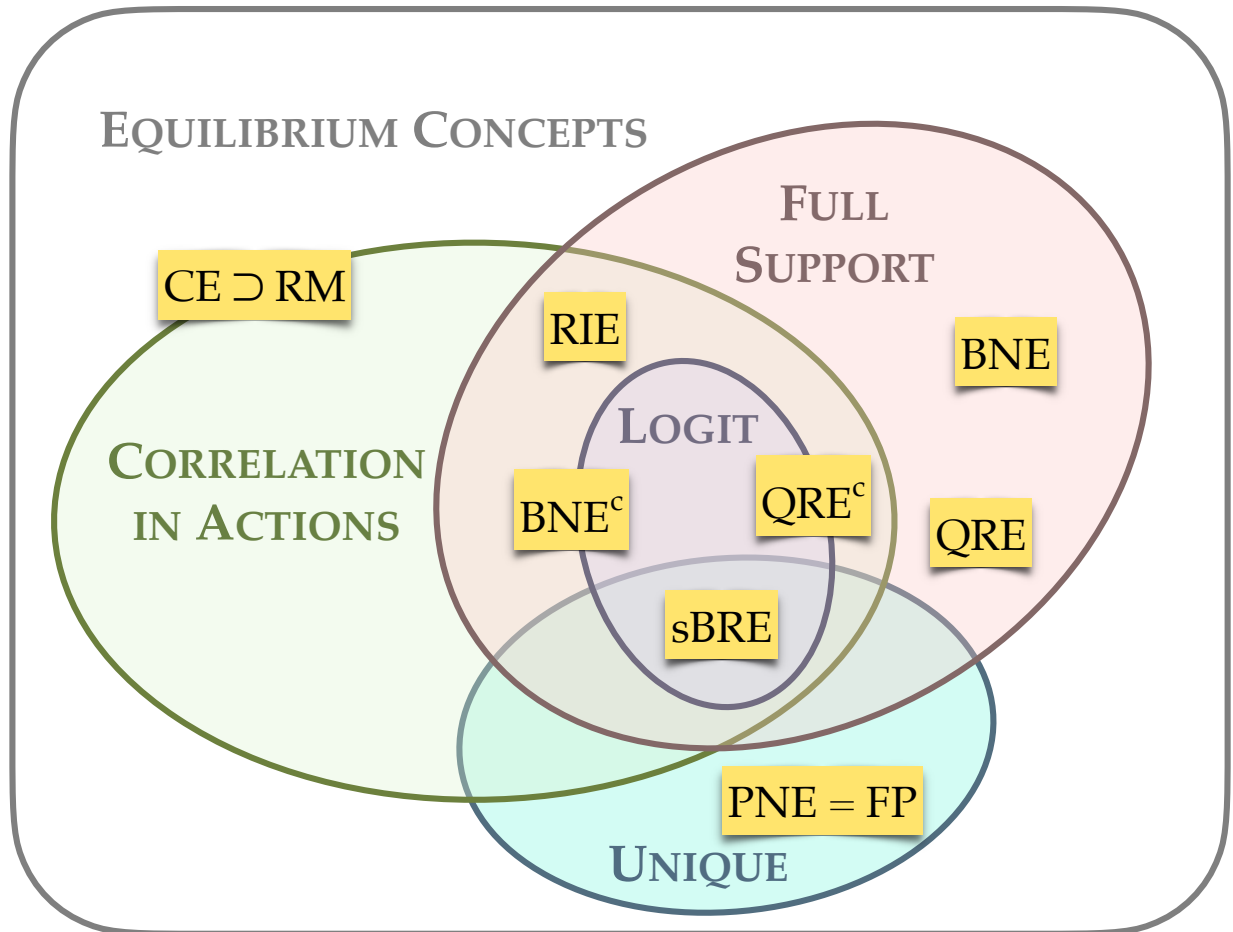


Figure 2: **Stochastic Best-Response Equilibrium in Context: the Venn Diagram.** $sBRE$ = stochastic best-response equilibrium, RIE = rational inattention equilibrium, QRE = quantile response equilibrium with independent mistakes, QRE^c = quantile response equilibrium with correlated mistakes, BNE = Bayes–Nash equilibrium with independent payoff shocks, BNE^c = Bayes–Nash equilibrium with correlated payoff shocks, CE = correlated equilibrium, RM = regret minimization, PNE = potential-maximizing Nash equilibrium, FP = fictitious play.

tion, and RIE’s inability to deliver the especially tractable functional form in (3) hampers estimation. Even when the cost of information acquisition is entropic, the equilibrium distribution—biased logit—lacks the supreme tractability of (3).

Correlation in participation decisions can be alternatively generated by replacing spying with a randomization device and a mediator, thereby invoking the correlated equilibrium (CE). CE shares in RIE’s problems, though, and has more. CEs are multiple. Some CEs exhibit no correlation in participation decisions. Among those that do, none may conform with the logit functional form in (3).

Related to CE is [Hart and Mas-Colell’s \(2000\)](#) regret-minimization (RM) protocol. The protocol differs from the best-response dynamics that motivates sBRE and always converges to a CE.

The existence of aforementioned alternatives (along with some other alternatives we discuss shortly) suggests that sBRE’s core feature—the correlation in participation decisions—is not fragile. It can emerge for a variety of reasons. Some common equilibrium concepts do resist correlation, though. In order to circumscribe the model’s scope, we briefly discuss these concepts.

Just like sBRE, [McKelvey and Palfrey’s \(1995\)](#) quantile-response equilibrium (QRE) assumes that agents make mistakes. As long as QRE maintains sBRE’s assumption that these mistakes are statistically independent, it rules out correlation in agents’ participation decisions. At QRE, agents best respond to the equilibrium probability distribution of others’ actions rather than to the actions themselves. There is no mechanism to correlate actions. Such a mechanism can be introduced, however, by correlating mistakes. Let QRE^c denote QRE with correlated mistakes. A joint probability distribution over all agents’ mistakes can be manufactured to make QRE^c ’s probability distribution over participation profiles match the sBRE (or any other probability distribution; see [Haile, Hortacsu and Kosenok 2008](#), Theorem 1). Neither QRE nor QRE^c need be unique. Full support, by contrast, arises naturally when mistakes are unbounded.

In our setting, the Bayes–Nash equilibrium (BNE) is formally equivalent to QRE if one interprets the stochasticity in agents’ actions as arising from payoff shocks rather than mistakes. As a result, BNE, too, rules out correlation in agents’ actions unless one assumes that payoff shocks are correlated.

The Nash equilibrium (NE) would seem a natural solution concept, but it predicts both too little and too much. NE predicts too little because platform participation is a coordination game, plagued by the multiplicity of NEs. The multiplicity problem admits a compelling solution, though. In two-sided market settings, [Chan \(2021\)](#) proposes to select the NE with the maximal potential. This refinement, denoted by PNE, is com-

elling because robust to the introduction of small amounts of incomplete information (Kajii and Morris, 1997; Ui, 2001). PNE is related to sBRE in that it is both sBRE’s modal prediction and sBRE’s limit as $\sigma \rightarrow 0$. NE predicts too much because it violates full support by assigning probability zero to most participation profiles, thereby forcing one to focus on the applications in which the data are not rich enough to identify model parameters. PNE shares in this shortcoming and bequeaths it upon fictitious play (FP), a type of best-response dynamics that converges to the (generically unique) PNE (Monderer and Shapley, 1996).

QRE^c and BNE^c each possess some of sBRE’s critical features: correlation, full support, and the logit distribution. Even so, we favor sBRE. While QRE^c and BNE^c directly assume how the joint probability distribution over agents’ mistakes or payoff shocks depends on the model parameters, sBRE treats mistakes as independent and instead derives the statistical dependence of agents’ participation decisions from payoffs in an intuitive way.

4 Restricted Boltzmann Machines: An Interlude

Conceived by Smolensky (1986), the restricted Boltzmann machine (RBM) is a parameterized probability distribution of vectors $\mathbf{x} \equiv (\mathbf{x}_1, \mathbf{x}_2)$ with values in $\{0, 1\}^n$. The distribution’s dependence structure is represented by a complete undirected bipartite graph. With its every node, we associate a binary random variable. Any two nodes on different sides of the graph—in different layers of the “neural net” the graph represents—are connected by an edge, meaning that the corresponding random variables are statistically dependent conditional on the values of the remaining variables. Any two nodes in the same layer are not connected by an edge, meaning that the corresponding random variables are statistically independent conditional on the values of the remaining variables. The strength of statistical dependence is not pinned down by the graph itself and must be assumed separately. RBM assumes that $\mathbf{x}$ is distributed according to the distribution in (3), where $\mathbf{W}$ and $\mathbf{b}$ are arbitrary parameters. That is, a probability distribution can be generated by some RBM if and only if it is the sBRE of some two-sided market.

In the context of machine learning, RBM is classified as a neural network. It strikes a balance between tractability and expressiveness. Early on, RBM proved successful at recognizing handwritten characters and recommending movies, among other tasks. Later on, RBM became a building block of more complex neural networks called deep Boltzmann machines (more on these later).

To accomplish a task, first, RBM must be trained. In most applications, only one layer of the RBM maps directly into data. This layer is said to be visible. For example, if the

task at hand is to recommend movies based on past user experience, then the visible layer comprises each user's preferences over movies: thumbs up or thumbs down for every movie. If the task is to recognize handwritten characters, then the visible layer lines up the pixels in the black-and-white pixelation of a handwritten character (on or off for each pixel) and the corresponding correct answers (for each handwritten character, whether it is 1, 2, or 3, etc.). In both applications, the other, hidden, layer comprises the latent variables whose sole purpose is to introduce a rich enough dependence structure among the visible variables. These latent variables do not directly correspond to anything observed in the data. The values of these variables are imputed in the course of training. Their interpretations are open-ended. In the movie-recommendation application, latent variables may correspond to genres and actors. In the character-recognition application, latent variables may correspond to strokes shared by multiple characters.

Training an RBM consists in choosing the underlying parameters ($\mathbf{W}, \mathbf{b}$) in such a way that the fitted distribution of the visible variables $\mathbf{x}_1$ generated by the RBM is close to the empirical distribution of $\mathbf{x}_1$ in the training data set. Training an RBM is analogous to estimating the parameters of a two-sided market when data are generated by sBRE with two exceptions:

1. In RBMs, only one layer is typically visible, whereas in two-sided markets, participation on both sides of the market is visible.
2. The goal of training RBMs is to fit the data, whereas the goal of estimating parameters of a two-sided market is to estimate the true parameter values of the underlying structural model.

The distinction between fit and estimation matters for the startup's second task: pricing. While a good fit helps predict agent participation at the prices at which the model has been trained, to price well, the startup must be able predict agent participation in a variety of counterfactual scenarios. Positive prices discourage participation, negative prices (i.e., subsidies) encourage it, and to predict the exact magnitude of these forces, one must know the underlying model parameters.

5 Learning the Two-Sided Market Parameters

The startup has access to T i.i.d. draws from the sBRE. These draws comprise the dataset, denoted by $(\mathbf{x}(t))_{t=1}^T$. We assume that these data are generated while the startup does not charge any prices. This assumption is inconsequential and is made to conserve notation.

This assumption is also realistic. Companies commonly offer access to a platform free of charge initially and seek to monetize the platform later. For instance, Upwork offered free access for both freelancers and clients from its inception in May 2015 through May 2019 (Upwork Inc., Form 10-K 2019, p. 58), whereas by July 2022, both freelancers and clients had to pay (Upwork Inc., Q2 2022 Shareholder Letter, p. 2).

5.1 Likelihood Maximization

We henceforth assume that the scale-of-noise parameter σ is known and focus on identifying the remaining model parameters $(\mathbf{W}, \mathbf{b})$. If the ultimate goal is to predict behavior, then the choice of σ is mere normalization of the units in which the components of $(\mathbf{W}, \mathbf{b})$ are measured. If the ultimate goal is to price optimally, then $(\mathbf{W}, \mathbf{b})$ must be compared to prices and, so, must be measured in dollars. In practical applications, to pin down the units, one can use the following stand-alone trick: Modify the two-sided market so that whenever, say, Alice and Bob both show up, instead of letting them interact, the platform tells them to stay away from each other and instead pays each of them a dollar. As a result, by construction, $W_{Alice,Bob} = \$1$. The estimates for $(\mathbf{W}, \mathbf{b})$ assuming an arbitrary σ are then scaled to respect $W_{Alice,Bob} = \$1$; the scaling factor reveals the underlying σ .

We henceforth maintain the technical assumption that the true parameter values $(\mathbf{W}, \mathbf{b})$ are confined to the interior of a certain compact and convex set, no matter how large.

Lemma 1 exhibits an estimator that is **consistent** in the sense of converging in probability to the true parameters $(\mathbf{W}, \mathbf{b})$ as the sample size T grows. As a result, the model parameters are identified.

Lemma 1. *The maximum-likelihood estimator*

$$(\mathbf{W}^{ML}, \mathbf{b}^{ML}) \equiv \arg \max_{\tilde{\mathbf{W}}, \tilde{\mathbf{b}}} L \equiv \frac{1}{T} \sum_{t=1}^T \log \mathbb{P}(\mathbf{x}(t) \mid \tilde{\mathbf{W}}, \tilde{\mathbf{b}})$$

with $\mathbb{P}$ defined in (3) is consistent for $(\mathbf{W}, \mathbf{b})$.

Proof. Consistency follows from the strict convexity of the log-likelihood function L . Strict concavity follows from Proposition 3.1 of [Wainwright and Jordan \(2008, p. 62\)](#) because sBRE is a member of the minimal exponential family. ■

The maximum likelihood estimator in Lemma 1 achieves identification by matching moments. Roughly speaking, Alice's standalone value from participation is identified off her empirical frequency of participation, and her benefit from interacting with Bob is

identified off the empirical correlation between the pair's participation decisions. Formally, this logic is captured by the first-order conditions evaluated at the maximum-likelihood (ML) estimates $(\mathbf{W}^{ML}, \mathbf{b}^{ML})$:

$$\begin{aligned} \sum_{\mathbf{y} \in \{0,1\}^n} \mathbf{y} \mathbb{P}(\mathbf{y} \mid \mathbf{W}^{ML}, \mathbf{b}^{ML}) &= \frac{1}{T} \sum_{t=1}^T \mathbf{x}(t) \\ \sum_{\mathbf{y} \in \{0,1\}^n} \mathbf{y}_1 \mathbf{y}_2^\top \mathbb{P}(\mathbf{y} \mid \mathbf{W}^{ML}, \mathbf{b}^{ML}) &= \frac{1}{T} \sum_{t=1}^T \mathbf{x}_1(t) \mathbf{x}_2(t)^\top. \end{aligned} \quad (4)$$

The left-hand sides in (4) are the moments of the estimated distribution and the right-hand sides are the corresponding moments in the data. System (4) admits no explicit solution.

Even computationally, likelihood maximization presents challenges. The likelihood function suffers from **combinatorial explosion**: in equation (3), the 2^n terms in the denominator $\sum_{\mathbf{y} \in \{0,1\}^n} \exp\left(\frac{1}{\sigma} (\mathbf{b}^\top \mathbf{y} + \mathbf{y}_1^\top \mathbf{W} \mathbf{y}_2)\right)$ are too many to compute even for moderate values of n . That is, computing the likelihood function—let alone maximizing it—is a numerically intractable task.

5.2 Contrastive Divergence

Contrastive divergence (CD) is a bastardization of likelihood maximization by gradient ascent. Its questionable pedigree notwithstanding, it turns out to be consistent (in the sense made precise below).

In order to introduce CD, let us start with gradient ascent for likelihood maximization. Let us focus on the gradient with respect to $\mathbf{b}$ evaluated at step k of gradient ascent when the current guess for parameter values is $(\mathbf{W}^{[k]}, \mathbf{b}^{[k]})$:

$$\nabla_{\mathbf{b}} L^{[k]} = \frac{1}{T} \sum_{t=1}^T \mathbf{x}(t) - \sum_{\mathbf{y} \in \{0,1\}^n} \mathbf{y} \mathbb{P}(\mathbf{y} \mid \mathbf{W}^{[k]}, \mathbf{b}^{[k]}). \quad (5)$$

The gradient $\nabla_{\mathbf{W}} L^{[k]}$ can be analyzed analogously. In (5), the first sum involves only data and is easy to compute. The second sum contains the intractable number 2^n of terms, each of which contains the intractable function $\mathbb{P}(\mathbf{y} \mid \mathbf{W}^{[k]}, \mathbf{b}^{[k]})$. This combinatorial explosion is due to the heterogeneity of network effects. When network effects are homogeneous (i.e., when all entries of $\mathbf{W}$ are the same), both the expression for the equilibrium prob-

ability $\mathbb{P}$ in (3) and the sum in the gradient (5) simplify, and the number of terms in the sums drops from 2^n down to $O(n^2)$.

Towards making progress on gradient ascent, note that the intractable sum in (5) is an expectation. Approximate this expectation with the tractable sum:

$$\nabla_{\mathbf{b}} L^{[k]} \approx \frac{1}{T} \sum_{t=1}^T \mathbf{x}(t) - \frac{1}{T} \sum_{t=1}^T \mathbf{x}^{(\infty)}(t), \quad (6)$$

where each $\mathbf{x}^{(\infty)}(t)$ is ideally an i.i.d. draw from the distribution $\mathbb{P}(\cdot \mid \mathbf{W}^{[k]}, \mathbf{b}^{[k]})$ whose unknown expectation we seek to approximate. We manufacture each draw $\mathbf{x}^{(\infty)}(t)$ by starting out at the t -th data point $\mathbf{x}(t)$ and then evolving it via a sequence of m stochastic best-response steps under the assumption that the model parameters are $(\mathbf{W}^{[k]}, \mathbf{b}^{[k]})$ to obtain $\mathbf{x}^{(m)}(t)$. By Proposition 1, as the number of stochastic best-response steps m goes to infinity, stochastic best-response dynamics culminates in a $\mathbf{x}^{(m)}(t)$ that in distribution converges to $\mathbf{x}^{(\infty)}$, a draw from $\mathbb{P}(\cdot \mid \mathbf{W}^{[k]}, \mathbf{b}^{[k]})$. Then, the average $\frac{1}{T} \sum_t \mathbf{x}^{(m)}(t)$ of the T draws so obtained approximates the sought expectation $\sum_{\mathbf{y}} \mathbf{y} \mathbb{P}(\mathbf{y} \mid \mathbf{W}', \mathbf{b}')$. In statistics, the described best-response dynamics until convergence in distribution goes by the name **MCMC**, or Markov chain Monte Carlo. Because convergence is in distribution only, the approximate gradient on the right-hand side of (6) is a random variable.

The problem with running MCMC for a “large” number m of steps at every iteration k of gradient ascent is that the process is impractically slow. This slowness is an empirical fact. Of course, one can speed up MCMC by taking m to be “small” (e.g., $m = 1$ or $m = 5$), but then convergence to a random draw from $\mathbb{P}(\cdot \mid \mathbf{W}^{[k]}, \mathbf{b}^{[k]})$ will no longer occur. This unprincipled act of despair—taking m to be “small”—has a name. It is called **contrastive divergence** (CD). While the moniker itself is whimsical, its existence is hopeful: someone must have found the unprincipled approach that invalidates the approximate equality in (6) to “work.” And work it does, in a sense that we now make precise.

Let CD- m denote contrastive divergence whose every chain of stochastic best-response dynamics described above is terminated after just m steps. CD- m amounts to “nongradient” ascent whereby, at every iteration k , instead of climbing the likelihood function by following the computationally intractable gradient (6), we attempt to climb the likelihood function by following the computationally tractable **nongradient**:

$$\Delta_{\mathbf{b}} L^{[k]} \equiv \frac{1}{T} \sum_{t=1}^T \mathbf{x}(t) - \frac{1}{T} \sum_{t=1}^T \mathbf{x}^{(m)}(t)$$

and its analogously defined counterpart $\Delta_{\mathbf{W}}L^{[k]}$ in full realization that $\Delta_{\mathbf{b}}L^{[k]} \not\approx \nabla_{\mathbf{b}}L^{[k]}$ and $\Delta_{\mathbf{W}}L^{[k]} \not\approx \nabla_{\mathbf{W}}L^{[k]}$. Let $(\mathbf{W}^{[k,m]}, \mathbf{b}^{[k,m]})$ denote the value of the CD- m estimator as described in the paragraph above after k iterations of nongradient ascent.

Proposition 2. *Contrastive divergence is consistent for $(\mathbf{W}, \mathbf{b})$ in the sense that there exists a bounded positive integer m such that, as the sample size T grows, the CD- m estimator*

$$\lim_{K \rightarrow \infty} \sum_{k=1}^K (\mathbf{W}^{[k,m]}, \mathbf{b}^{[k,m]})$$

converges in probability to $(\mathbf{W}, \mathbf{b})$.

The technical nature of the proof of Proposition 2 is responsible for its relegation to Appendix A.2. Our proof relies on Jiang, Wu, Jin and Wong’s (2018) exacting study of CD in the models in the canonical exponential family. We show that our model belongs to this family and, therefore, CD is consistent if the assumptions in Jiang, Wu, Jin and Wong’s Theorem 2.1 can be verified. We verify these assumptions by following the roadmap in Jiang, Wu, Jin and Wong’s example (their Section 4.2) of a fully visible RBM with $n_1 = n_2 = 1$.

Even though motivated by the problem of training RBMs, Jiang, Wu, Jin and Wong’s (2018) consistency result does not cover the vast majority of RBMs, which have a hidden layer. Our model is covered because both sides of the market are assumed to be visible.

5.3 The Comparative Tractability of Contrastive Divergence

While CD’s consistency is a formal proposition, its numerical tractability is an empirical fact. This fact is suggested by two decades of neural-network training, during which practitioners would brazenly set the “bounded positive integer m ” of Proposition 2 to 1 and proceed to train RBMs fast and effectively. This fact is also corroborated by our simulations, to which we now turn. All simulations have been conducted using Google Colab Pro, a cloud-based platform that provides access to high-performance computing resources. We used the GPU (Graphics Processing Unit) accelerator NVIDIA Tesla T4, which is optimized for machine-learning tasks. For transparency and reproducibility, our code is available in the Google Colab notebook at [this link](#).

In each simulation, we first draw the underlying “true” parameters $(\mathbf{W}, \mathbf{b})$ from the multivariate standard normal distribution. We then synthesize a fictitious dataset by drawing T participation profiles from the sBRE distribution induced by the parameters $(\mathbf{W}, \mathbf{b})$ and $\sigma = 1$. We then see how capable CD and likelihood maximization (ML) are of

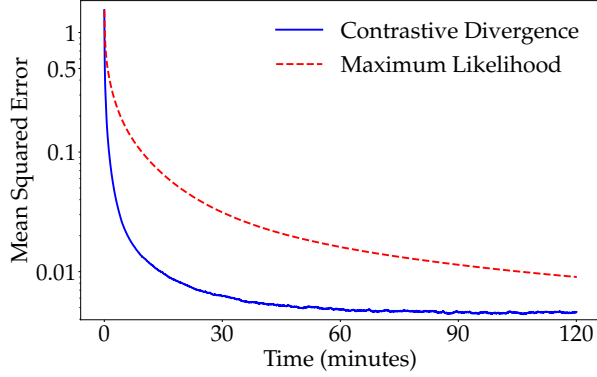


Figure 3: **With 11 agents on each side, conventional likelihood maximization is slow, whereas contrastive divergence is fast.** The underlying simulated data set has $T = 100,000$ observations.

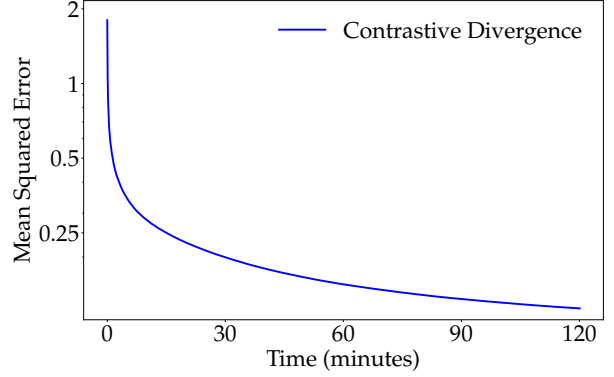


Figure 4: **With 15 agents on each side, contrastive divergence continues to converge.** The underlying simulated data set has $T = 500,000$ observations.

uncovering $(\mathbf{W}, \mathbf{b})$ under the assumption that $\sigma = 1$ by reporting each estimate’s mean squared error in the course of each algorithm’s execution. Here, CD stands for CD- m with $m = 1$ under blocked sampling, and ML stands for likelihood maximization under gradient ascent with each $2^{n_1+n_2}$ -term sum computed using vectorization. Both CD and ML are initialized from the same random guess $(\mathbf{W}^{[0]}, \mathbf{b}^{[0]})$.

Figure 3 illustrates how much faster CD converges to $(\mathbf{W}, \mathbf{b})$ than ML does in a market with $n_1 = n_2 = 11$ agents on each side. The figure shows the mean squared error of each estimator as a function of time elapsed in the simulation. Because CD is inherently stochastic, the corresponding curve is jagged. What converges in Proposition 2 is the moving average of the estimates that generate this curve.

Figure 4 drops ML and focuses on the performance of CD in a larger market, with $n_1 = n_2 = 15$ agents on each side. CD continues to converge, within a “reasonable” time frame and with “few” observations. For the remainder, let us focus on the number of observations. What is remarkable about this market is that, while it has $2^{n_1+n_2} \approx 1,000,000,000$ possible participation profiles, only $T = 500,000$ (possibly non-distinct) profiles suffice to estimate the model parameters. We say that $T = 500,000$ is not many because orders of magnitude more observations would have been required to deal with the pesky denominator in (3) by using the differencing approach commonly deployed in estimation of (smallish) discrete-choice models. Here is why.

To illustrate the differencing approach, let $\mathbf{x}$ and $\mathbf{z}$ be two participation profiles such that $\mathbf{z} = \mathbf{0}$ and $\mathbf{x}$ is zero everywhere except at $x_{1i} = 1$. Taking the difference of the loga-

rithms of the probabilities of these profiles at the sBRE eliminates the pesky denominator in (3) to yield:

$$\log \mathbb{P}(\mathbf{x} \mid \mathbf{b}, \mathbf{W}) - \log \mathbb{P}(\mathbf{z} \mid \mathbf{b}, \mathbf{W}) = \mathbf{b}^\top (\mathbf{x} - \mathbf{z}) + \mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2 - \mathbf{z}_1^\top \mathbf{W} \mathbf{z}_2,$$

whence

$$b_{1i} = \log \frac{\mathbb{P}(\mathbf{x} \mid \mathbf{b}, \mathbf{W})}{\mathbb{P}(\mathbf{z} \mid \mathbf{b}, \mathbf{W})}.$$

As a result, b_{1i} can be estimated as the logarithm of the ratio of the observed frequencies of $\mathbf{x}$ and $\mathbf{z}$. To estimate b_{1i} so with any degree of precision, the profiles $\mathbf{x}$ and $\mathbf{z}$ —just two profiles out of the $2^{n_1+n_2}$ possible ones—must be observed over and over again, thereby requiring a data set of a size $T \gg 2^{n_1+n_2}$. This requirement contrasts with the simulation above, where $T \ll 2^{n_1+n_2}$ observations suffice because CD (just like ML) does not toss away any: every observation contains information that is directly or indirectly relevant for estimating b_{1i} (and other parameters).

6 Pricing

This section discusses optimal pricing in the special case in which σ , the scale-of-noise parameter, is “small.” This special case happens to be the focus of the existing literature on platform pricing. The section carries the hopeful message that estimation has not been in vain. The parameter estimates can be put to use in service of welfare or revenue.

6.1 A Class of Pricing Schemes

We consider the most general pricing scheme that preserves the strategic structure of the participation game. This pricing scheme has two components. The first component is a matrix $\mathbf{P} \equiv (P_{ij})_{i,j} \in \mathbb{R}^{n_1 \times n_2}$ of interaction prices. When agent i on side 1 and agent j on side 2 both join the platform, each pays the amount P_{ij} . The second component is a vector $\mathbf{p} \equiv (\mathbf{p}_1, \mathbf{p}_2) \in \mathbb{R}^n$ of personalized access prices, where $\mathbf{p}_1$ and $\mathbf{p}_2$ are the price vectors faced by the agents on sides 1 and 2 of the market, respectively. As a result, at a participation profile $\mathbf{x}$, agents on the sides 1 and 2 of the market end up paying their corresponding entries in the respective payment vectors

$$\mathbf{p}_1 + \mathbf{P} \mathbf{x}_2 \quad \text{and} \quad \mathbf{p}_2 + \mathbf{P}^\top \mathbf{x}_1.$$

The equilibrium probability of a participation profile $\mathbf{x}$ when prices are $(\mathbf{P}, \mathbf{p})$ is $\mathbb{P}(\mathbf{x} \mid \mathbf{W} - \mathbf{P}, \mathbf{b} - \mathbf{p})$, which is the obvious adjustment of the corresponding probability (3).

6.2 Pricing for Welfare

We lead with welfare maximization, and we do so for two reasons: conceptual clarity and realism. Conceptually, revenue maximization in our setting requires the platform to maximize welfare before extracting it. Realistically, early in its existence, a platform wants to maximize welfare in order to attract new customers and achieve market dominance, all the while passing under the radar of antitrust authorities, who have traditionally been blind to welfare-maximizing pricing (Bork, 1978). For instance, in 2015, Uber suffered a loss of about \$2 billion while generating the consumer surplus of \$6.8 billion across the U.S. cities (Cohen, Hahn, Hall, Levitt and Metcalfe, 2016). The first year in which Uber turned a profit was 2023.

The Problem

The welfare maximization problem takes the form

$$\max_{\mathbf{P}, \mathbf{p}} \sum_{\mathbf{x} \in \{0,1\}^n} \mathbb{P}(\mathbf{x} \mid \mathbf{W} - \mathbf{P}, \mathbf{b} - \mathbf{p}) \left(\mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2 \right), \quad (7)$$

where $\mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2$ is utilitarian welfare, the sum of the platform's revenue and the agents' payoffs. Utilitarian welfare counts bilateral benefits from interaction twice because they accrue twice: once to each agent in the interacting pair. The welfare does not contain prices, which are a wash between the platform and the agents. Nor does the welfare contain any epsilons that appear in the definition of stochastic best response in (2); our interpretation is that these epsilons encode mistakes rather than utility shocks.

The problem in (7) suffers from combinatorial explosion because of 2^n terms in the sum and 2^n terms inside each probability $\mathbb{P}(\mathbf{x} \mid \mathbf{W} - \mathbf{P}, \mathbf{b} - \mathbf{p})$. Here, however, CD is of no avail. And yet, we plough ahead.

The "Price of Anarchy"

Each agent's participation decision imposes an externality on agents on the other side of the market. Because no agent internalizes this externality, equilibrium is generally

inefficient in the sense of falling short of maximizing the **first-best welfare**, defined as

$$\max_{\mathbf{x} \in \{0,1\}^n} \left\{ \mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2 \right\}.$$

Proposition 3 (proved in Appendix A.3) shows that equilibrium welfare can fall short of the first-best by quite a lot. As a result, there is a problem for pricing to solve, or at least to mitigate.

Proposition 3 (The Price of Anarchy). *With no corrective pricing (i.e., under $\mathbf{P} = \mathbf{0}$ and $\mathbf{p} = \mathbf{0}$), equilibrium welfare can be an arbitrarily small fraction of the first-best welfare. In other words, the “price of anarchy” is infinite.*

The idea behind Proposition 3 is to conceive of economies in which network effects are strong enough for welfare to be maximized when all participate but not strong enough for anyone to actually want to participate. This discrepancy between welfare-maximizing behavior and equilibrium behavior arises because participants do not internalize the externalities.

Welfare-Maximizing Prices when $\sigma \rightarrow 0$

In the special case when $\sigma \rightarrow 0$ (the mistakes in best responses vanish), the welfare-maximization problem in (7) admits a simple solution. This special case is interesting because the conclusion of Proposition 3 continues to hold even when $\sigma \rightarrow 0$, as is evident in the proposition’s proof.

Proposition 4 (Welfare-Maximizing Prices). *If $\mathbf{P} = (1 - \gamma) \mathbf{W}$ and $\mathbf{p} = \left(1 - \frac{1}{2}\gamma\right) \mathbf{b}$ for an arbitrary positive γ , then equilibrium welfare converges to the first-best welfare as $\sigma \rightarrow 0$.*

Proof. Under the prices in the proposition’s statement, expected welfare satisfies

$$\lim_{\sigma \rightarrow 0} \frac{\sum_{\mathbf{x} \in \{0,1\}^n} \exp\left(\frac{\gamma}{2\sigma} [\mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2]\right) [\mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2]}{\sum_{\mathbf{x} \in \{0,1\}^n} \exp\left(\frac{\gamma}{2\sigma} [\mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2]\right)} = \max_{\mathbf{x} \in \{0,1\}^n} \left\{ \mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2 \right\},$$

as desired. ■

When $\gamma = 1$, Proposition 4 prescribes $\mathbf{P} = \mathbf{0}$ and $\mathbf{p} = \frac{1}{2}\mathbf{b}$, thereby cutting each agent’s standalone value from joining in half while leaving the experienced interaction benefits unchanged. As a result, the relative salience of interaction benefits doubles. This doubling induces each agent to internalize the externality from his participation. An alternative way to achieve the same goal is to set $\gamma = 2$ and, so, charge $\mathbf{P} = -\mathbf{W}$ (Pigouvian

interaction subsidies) and $\mathbf{p} = \mathbf{0}$ (no access fees). Which value of γ is appropriate depends on whether revenue is a desideratum or a liability, and on whether nonzero interaction prices are practical.

6.3 Pricing for Revenue

Profit—here, synonymous with revenue—is the platform’s ultimate reward.

The Problem

The expected-revenue maximization problem takes the form

$$\max_{\mathbf{P}, \mathbf{p}} \sum_{\mathbf{x} \in \{0,1\}^n} \mathbb{P}(\mathbf{x} \mid \mathbf{W} - \mathbf{P}, \mathbf{b} - \mathbf{p}) \left(\mathbf{p}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{P} \mathbf{x}_2 \right), \quad (8)$$

where $\mathbf{p}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{P} \mathbf{x}_2$ is the platform’s revenue. The revenue is the sum of access fees $\mathbf{p}^\top \mathbf{x}$ and the interaction fees $2\mathbf{x}_1^\top \mathbf{P} \mathbf{x}_2$. Each interaction fee must be counted twice because it is collected from each member of the participating pair. From the perspective of revenue, it does not matter whether the epsilons in the best-response equation (2) are interpreted as mistakes or as shocks to payoffs.

The Price of Model Misspecification

Just like refraining from corrective pricing may come at a great cost as far as welfare is concerned (Proposition 3), the failure to account for heterogeneous network effects in learning and pricing may come at a great cost as far as revenue is concerned (Proposition 5, proved in Appendix A.4). Should the platform ignore the heterogeneity of network effects, it will make wrong inferences about the underlying model parameters and will make wrong pricing decisions. In fact, pricing based on the misspecified model may be arbitrarily ruinous: the revenue may be arbitrarily low relative to the revenue the platform would have achieved had it used the correct model.

Proposition 5 (The Price of Model Misspecification). *The maximal revenue under the misspecified model that (wrongly) assumes homogeneous network effects can be an arbitrarily small fraction of the optimal revenue.*

Proposition 5 exploits the idea that if benefits from interaction are positive for some pairs of agents and negative for others, then so is the correlation in the corresponding agents’ equilibrium participation decisions. The platform that knows that the true model

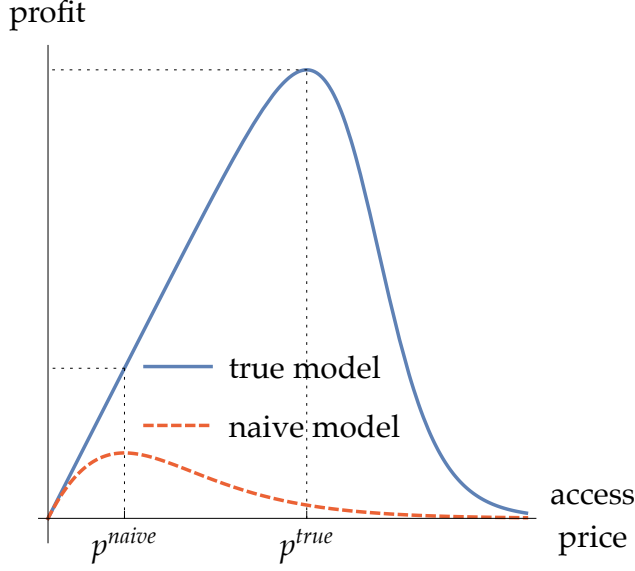


Figure 5: **Ignoring heterogeneous network effects lowers revenue.** The naive platform, which wrongly assumes that network effects are homogeneous, prices access at p^{naive} , thereby inducing lower revenue under the true model than the optimal access price p^{true} would.

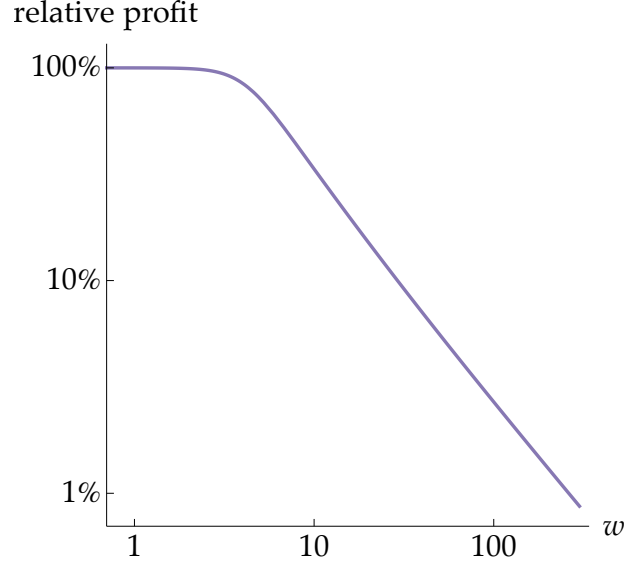


Figure 6: **The stronger the neglected network effects (the larger w), the lower is the fraction of the optimal revenue that the naive platform extracts.**

admits heterogeneous network effects exploits this correlation to correctly identify the underlying parameters. The platform that naively believes the network effects to be homogeneous only looks at aggregate behavior. In the aggregate, positive and negative correlations cancel out, and, viewed through the lens of the misspecified model, all network effects appear to be zero. As a result, the naive platform does not even attempt to extract the value of network effects. Figure 5 illustrates this mispricing in an example in which network effects are w for some pairs of agents and $-w$ for others. Figure 6 illustrates that this mispricing costs the naive platform arbitrarily dearly when w is arbitrarily large. The details of the underlying example are in the proposition's proof.

Proposition 5 is the *raison d'être* for this paper.

Revenue-Maximizing Prices when $\sigma \rightarrow 0$

In the special case in which $\sigma \rightarrow 0$, the revenue-maximization problem in (8) admits a simple solution. This special case is still interesting because the conclusion of Proposi-

tion 5 continues to hold when $\sigma \rightarrow 0$, as can be checked by consulting the proposition's proof.

Proposition 6 (Revenue-Maximizing Prices). *If $\mathbf{P} = (1 - \sqrt{\sigma}) \mathbf{W}$ and $\mathbf{p} = (1 - \frac{1}{2}\sqrt{\sigma}) \mathbf{b}$, then equilibrium revenue converges to its upper bound, the first-best welfare, as $\sigma \rightarrow 0$.*

Proof. At the specified prices, the expected revenue satisfies

$$\lim_{\sigma \rightarrow 0} \frac{\sum_{\mathbf{x} \in \{0,1\}^n} \exp\left(\frac{1}{2\sqrt{\sigma}} [\mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2]\right) (\mathbf{p}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{P} \mathbf{x}_2)}{\sum_{\mathbf{y} \in \{0,1\}^n} \exp\left(\frac{1}{2\sqrt{\sigma}} [\mathbf{b}^\top \mathbf{y} + 2\mathbf{y}_1^\top \mathbf{W} \mathbf{y}_2]\right)} = \max_{\mathbf{x} \in \{0,1\}^n} \left\{ \mathbf{b}^\top \mathbf{x} + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2 \right\},$$

which is the first-best welfare. As $\sigma \rightarrow 0$, the lower bound on each agent's expected payoff converges to zero. Therefore, the upper bound on the platform's revenue converges to the first-best welfare. $\blacksquare$

Because the prices in Proposition 6 collect the first-best welfare in revenue, they must generate the first-best welfare. Indeed, the prices correspond to the welfare-maximizing prices of Proposition 4 with $\gamma = \sqrt{\sigma}$.

Full extraction stands in contrast to the divide-and-conquer pricing schemes whose optimality has been documented in the existing literature with homogeneous network effects (Chan, 2021). These schemes extract the entire surplus from some agents ("conquer") and take nothing from others ("divide"). (A typical example is access to a night club: men pay an arm and a leg, whereas women get in free.) Our full-extraction result relies on the feasibility of nonzero interaction prices. It does not rely on the hitherto maintained assumption that the value from each interaction is split equally. We go on record with this assertion in Remark 1, made precise and proved in Appendix A.5.

Remark 1. In the model's extension in which all agents on one side of the market systematically capture a higher fraction of the benefit from bilateral interaction than all agents on the other side do, the full-extraction result in Proposition 6 continues to hold.

7 Concluding Remarks: The Scope of the Analysis

The paper's analysis reaches beyond applications to monopolistic two-sided markets. Figure 7 announces the three extensions we consider below.

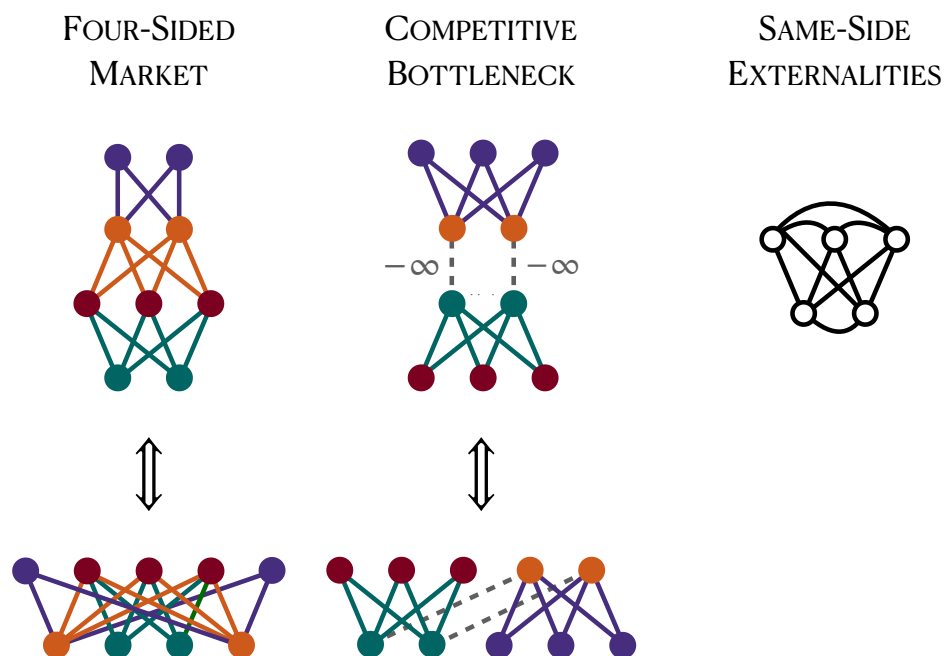


Figure 7: **Three extensions.** Multi-sided markets (left) and competitive bottlenecks (center) are formally equivalent to two-sided markets. Markets with same-side externalities (right) are not two-sided.

Multi-Sided Markets

Our two-sided markets framework accommodates without modification a class of multi-sided markets. Consider the following instance of a four-sided market. A professional-networking platform has four sides: advertisers, professionals, recruiters, and companies. Agents on the same side do not interact. Across sides, advertisers (side 1) interact with professionals (side 2), who also interact with recruiters (side 3), who also interact with companies (side 4). In other words, the sides are **stacked**: only the groups with consecutive indices interact. Let $\mathbf{W}_{12}$, $\mathbf{W}_{32}$, and $\mathbf{W}_{34}$ denote the matrices of interaction effects between groups 1 and 2, 3 and 2, and 3 and 4, respectively. Let $\mathbf{b}$ denote the vector of standalone values from participation, as before. The participation game can be verified to be a potential game with the potential

$$\mathbf{b}^\top \mathbf{x} + \mathbf{x}_1^\top \mathbf{W}_{12} \mathbf{x}_2 + \mathbf{x}_3^\top \mathbf{W}_{32} \mathbf{x}_2 + \mathbf{x}_3^\top \mathbf{W}_{34} \mathbf{x}_4 = \mathbf{b}^\top \mathbf{x} + \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_3 \end{pmatrix}^\top \begin{pmatrix} \mathbf{W}_{12} & \mathbf{0} \\ \mathbf{W}_{32} & \mathbf{W}_{34} \end{pmatrix} \begin{pmatrix} \mathbf{x}_2 \\ \mathbf{x}_4 \end{pmatrix}.$$

The right-hand side of the equation in the display above reveals that the four-sided market is formally equivalent to the two-sided market that treats sides 1 and 3 as one side and sides 2 and 4 as the other side. The (bidiagonal) block matrix collects network effects. This reduction to two-sided markets holds for any number of sides in any stacked multi-sided market. This reduction is only possible because our two-sided market model admits heterogeneous network effects.

The reduction implies that all of the results considered so far carry over. CD is fast and consistent. CD estimates can be used to price as discussed.

Competitive Bottlenecks

A **competitive bottleneck** is the name attached by [Armstrong \(2006\)](#) to the situation in which two two-sided platforms compete with each other for participants, but only on one side. Ride-sharing makes for a good stylized example. Uber and Lyft, each a two-sided market, compete for riders but not for drivers. The motivation is that each rider installs and uses both Uber and Lyft apps, whereas drivers are assumed to be locked into either Uber or Lyft with quests, bonuses, and computer game-like incentives designed by the scheming platforms.

A competitive bottleneck is formally equivalent to a four-sided market. The four sides are Uber drivers (side 1), Lyft drivers (side 4), and the riders each of whom is split into two selves: the Uber-contemplating self (side 2) and the Lyft-contemplating self (side 3).

Uber drivers transact with Uber-contemplating riders, Lyft drivers transact with Lyft-contemplating riders, and the two selves of each rider transact with each other. The fictitious split of each rider into his Uber and Lyft selves is a modeling device that enables each rider to choose which ride-sharing service to use, if any, while precluding the rider from choosing both (i.e., hopping into an Uber and a Lyft simultaneously). The described constraint on riders' choices is modeled by setting $\mathbf{W}_{32} \equiv -\infty \cdot \mathbf{I}$, where $\mathbf{I}$ is the identity matrix and $\infty \cdot 0 = 0$ by convention. The four-sided market that is the competitive bottleneck then reduces to a two-sided market with the potential

$$\mathbf{b}^\top \mathbf{x} + \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_3 \end{pmatrix}^\top \begin{pmatrix} \mathbf{W}_{12} & \mathbf{0} \\ -\infty \cdot \mathbf{I} & \mathbf{W}_{34} \end{pmatrix} \begin{pmatrix} \mathbf{x}_2 \\ \mathbf{x}_4 \end{pmatrix},$$

as one can check.

The reduction to a two-sided market once again ensures that CD is fast and consistent. Pricing for welfare is unchanged. Pricing for revenue is now interpreted as the collusive pricing solution. Duopolistic pricing is an open problem.

Same-Side Externalities

Same-side externalities (or interaction effects) occur when agents care about participation decisions of others on the same side of the market. Same-side externalities open up a whole new array of applications, such as search models of unemployment à la Mortensen–Pissarides (Pissarides, 2000), wherein job seekers exert congestion externalities on other job seekers, and vacancies posted by one firm exert congestion externalities on vacancies posted by other firms. Let matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ encode same-side externalities for sides 1 and 2 of the market, respectively. These matrices have zeros along the diagonals; no agent exerts an externality on himself. The potential can be verified to be

$$\mathbf{b}^\top \mathbf{x} + \mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2 + \mathbf{x}_1^\top \mathbf{S}_1 \mathbf{x}_1 + \mathbf{x}_2^\top \mathbf{S}_2 \mathbf{x}_2.$$

There is nothing two-sided about two-sided markets with same-side externalities. In machine learning, these markets correspond to (unrestricted) **Boltzmann machines**. Nonetheless, the equilibrium characterization in Proposition 1 continues to hold, provided in the definition of the sBRE attention is restricted to the revision protocols in which no two agents revise their action simultaneously, and the potential in (3) has been replaced by the expression in the display above. CD remains consistent, although it is now slower. The reason for the slowdown is the impossibility of using blocked sampling.

A Appendix: Omitted Proofs

A.1 Proof of Proposition 1

The sequence of participation profiles generated by the stochastic best-response dynamics in sBRE is a Markov chain. This chain is irreducible and aperiodic. The ergodic theorem assures existence of a unique stationary distribution for this chain. This distribution's functional form, in (3), is the concern for the rest of the proof.

The proof strategy is to verify that the distribution $\mathbb{P}$ in (3)—where we have suppressed the dependence of $\mathbb{P}$ on $(\mathbf{W}, \mathbf{b})$ —is stationary by verifying a sufficient condition. In order to formulate this condition, let $\mathbb{T}(\mathbf{x}, \mathbf{z})$ denote the probability that a single step of best-response dynamics transforms a participation profile $\mathbf{x}$ into $\mathbf{z}$. The **detailed balance condition** (Bremaud, 2020, Corollary 2.4.15, p. 92) verifies that $\mathbb{P}$ is the stationary distribution if, for all $\mathbf{x}$ and all $\mathbf{z}$, we have

$$\mathbb{P}(\mathbf{x}) \mathbb{T}(\mathbf{x}, \mathbf{z}) = \mathbb{P}(\mathbf{z}) \mathbb{T}(\mathbf{z}, \mathbf{x}). \quad (\text{A.1})$$

The condition says that $\mathbb{P}$ is a stationary distribution if the “outflow” of probability from every state $\mathbf{x}$ to every state $\mathbf{z}$ reachable from $\mathbf{x}$ in one step of best-response dynamics equals the “inflow” of probability from $\mathbf{z}$ to $\mathbf{x}$; in other words, the two flows are exactly “balanced.”

The critical feature of any revision protocol in sBRE is that, at each step, actions on at most one side of the market are revised. Henceforth, we focus on the step at which actions on side 1 are (possibly) revised while actions on side 2 (definitely) remain fixed: $\mathbf{z}_2 = \mathbf{x}_2$. (The complementary case of side-2 revisions is analogous.)

Let $\mathbf{u} \in \{0, 1\}^{n_1}$ represent the side-1 agents who are called upon to revise their actions at the best-response step under consideration, with $u_i = 1$ corresponding to agent i being called upon, and $u_i = 0$ corresponding to him not being called upon. Assume that $\mathbf{u}$ is distributed according to some probability distribution μ that is the same at every step of best-response dynamics. Note that

$$\mu \left(\prod_{i: x_{1i} \neq z_{1i}} u_i = 1 \right) = 0 \quad \implies \quad \mathbb{T}(\mathbf{x}, \mathbf{z}) = \mathbb{T}(\mathbf{z}, \mathbf{x}) = 0 \quad \implies \quad (\text{A.1}) \text{ holds.}$$

That is, whenever at least one agent whose actions in $\mathbf{x}$ and $\mathbf{z}$ differ is given no chance to revise his actions, there is no way to transition from $\mathbf{x}$ to $\mathbf{z}$ or from $\mathbf{z}$ to $\mathbf{x}$ in one step of the revision process. As a result, in the remainder of the proof, we focus on the complemen-

tary case in which $\mathbf{x}$ and $\mathbf{z}$ are such that

$$\mu \left(\prod_{i: x_{1i} \neq z_{1i}} u_i = 1 \right) > 0. \quad (\text{A.2})$$

Let $\pi_i(z_{1i} \mid \mathbf{x}_2)$ denote the probability that the agent i who is called upon to revise his action by best-responding to the previous-step action profile $\mathbf{x}$ opts for an action $z_{1i} \in \{0, 1\}$. This probability depends on $\mathbf{x}$ only through $\mathbf{x}_2$ thanks to the bipartite structure of the two-sided market. Because ε_{1i} is distributed logistically, $\pi_i(z_{1i} \mid \mathbf{x}_2)$ satisfies

$$\pi_i(z_{1i} \mid \mathbf{x}_2) = \frac{e^{z_{1i}(b_{1i} + \mathbf{W}_{i\bullet}^\top \mathbf{x}_2)}}{1 + e^{b_{1i} + \mathbf{W}_{i\bullet}^\top \mathbf{x}_2}}, \quad (\text{A.3})$$

where we have innocuously set $\sigma = 1$ to carry less notation; $\sigma = 1$ stays for the rest of the proof. Because all $(\varepsilon_{1i})_{i=1}^{n_1}$ are independent, the transition probability $\mathbb{T}$ takes the form

$$\mathbb{T}(\mathbf{x}, \mathbf{z}) = \sum_{\mathbf{u} \in \{0,1\}^n} \mu(\mathbf{u}) \prod_{i=1}^{n_1} \left(u_i \pi_i(z_{1i} \mid \mathbf{x}_2) + (1 - u_i) \mathbf{1}_{\{z_{1i}=x_{1i}\}} \right). \quad (\text{A.4})$$

Using the definitions of $\mathbb{T}$ in (A.4) and $\mathbb{P}$ in (3) and $\mathbf{z}_2 = \mathbf{x}_2$, the detailed balance condition in (A.1) can be written as

$$\begin{aligned} & \frac{e^{\mathbf{b}_1^\top \mathbf{x}_1 + \mathbf{b}_2^\top \mathbf{x}_2 + \mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2}}{\sum_{\mathbf{y}} e^{\mathbf{b}^\top \mathbf{y} + \mathbf{y}_1^\top \mathbf{W} \mathbf{y}_2}} \sum_{\mathbf{u}} \mu(\mathbf{u}) \prod_{i=1}^{n_1} \left(u_i \pi_i(z_{1i} \mid \mathbf{x}_2) + (1 - u_i) \mathbf{1}_{\{z_{1i}=x_{1i}\}} \right) \\ &= \frac{e^{\mathbf{b}_1^\top \mathbf{z}_1 + \mathbf{b}_2^\top \mathbf{x}_2 + \mathbf{z}_1^\top \mathbf{W} \mathbf{x}_2}}{\sum_{\mathbf{y}} e^{\mathbf{b}^\top \mathbf{y} + \mathbf{y}_1^\top \mathbf{W} \mathbf{y}_2}} \sum_{\mathbf{u}} \mu(\mathbf{u}) \prod_{i=1}^{n_1} \left(u_i \pi_i(x_{1i} \mid \mathbf{x}_2) + (1 - u_i) \mathbf{1}_{\{z_{1i}=x_{1i}\}} \right). \end{aligned}$$

Cancelling the normalizing constant in the denominator and simplifying gives

$$\frac{\sum_{\mathbf{u}} \mu(\mathbf{u}) \prod_{i=1}^{n_1} \left(u_i \pi_i(x_{1i} \mid \mathbf{x}_2) + (1 - u_i) \mathbf{1}_{\{z_{1i}=x_{1i}\}} \right)}{\sum_{\mathbf{u}} \mu(\mathbf{u}) \prod_{i=1}^{n_1} \left(u_i \pi_i(z_{1i} \mid \mathbf{x}_2) + (1 - u_i) \mathbf{1}_{\{z_{1i}=x_{1i}\}} \right)} = e^{\mathbf{b}_1^\top (\mathbf{x}_1 - \mathbf{z}_1) + (\mathbf{x}_1 - \mathbf{z}_1)^\top \mathbf{W} \mathbf{x}_2}. \quad (\text{A.5})$$

For some partition $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ of the index set $\{1, 2, \dots, n_1\}$, we have $x_{1i} = 1$ and $z_{1i} = 0$ for all $i \in \mathcal{A}$, $x_{1i} = 0$ and $z_{1i} = 1$ for all $i \in \mathcal{B}$, and $x_{1i} = z_{1i}$ for all $i \in \mathcal{C}$. With this notation, the left-hand side of (A.5) becomes

$$\frac{\sum_{\mathbf{u}} \mu(\mathbf{u}) U(\mathbf{u}) \prod_{i \in \mathcal{A}} u_i \pi_i(1 \mid \mathbf{x}_2) \prod_{i \in \mathcal{B}} u_i \pi_i(0 \mid \mathbf{x}_2)}{\sum_{\mathbf{u}} \mu(\mathbf{u}) U(\mathbf{u}) \prod_{i \in \mathcal{A}} u_i \pi_i(0 \mid \mathbf{x}_2) \prod_{i \in \mathcal{B}} u_i \pi_i(1 \mid \mathbf{x}_2)},$$

where $U(\mathbf{u}) \equiv \prod_{i \in \mathcal{C}} (u_i \pi_i(x_{1i} \mid \mathbf{x}_2) + 1 - u_i)$. Substituting the definition of π_i from (A.3) into the display above gives

$$\begin{aligned}
& \frac{\sum_{\mathbf{u}} \mu(\mathbf{u}) U(\mathbf{u}) \prod_{i \in \mathcal{A}} \frac{u_i e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}}{1 + e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}} \prod_{i \in \mathcal{B}} \frac{u_i}{1 + e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}}}{\sum_{\mathbf{u}} \mu(\mathbf{u}) U(\mathbf{u}) \prod_{i \in \mathcal{A}} \frac{u_i}{1 + e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}} \prod_{i \in \mathcal{B}} \frac{u_i e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}}{1 + e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}}} \\
&= \frac{\sum_{\mathbf{u}} \mu(\mathbf{u}) U(\mathbf{u}) \prod_{i \in \mathcal{A} \cup \mathcal{B}} \frac{u_i}{1 + e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}} \prod_{i \in \mathcal{A}} e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}}{\sum_{\mathbf{u}} \mu(\mathbf{u}) U(\mathbf{u}) \prod_{i \in \mathcal{A} \cup \mathcal{B}} \frac{u_i}{1 + e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}} \prod_{i \in \mathcal{B}} e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}} \\
&= \frac{\prod_{i \in \mathcal{A}} e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}}{\prod_{i \in \mathcal{B}} e^{b_{1i} + \mathbf{w}_{i\bullet}^\top \mathbf{x}_2}},
\end{aligned}$$

where the last equality relies on the cancellation that can be carried out thanks to the assumption in (A.2). The right-hand side in the display above equals the right-hand side in the detailed balance condition (A.5), which is thereby verified.

A.2 Proof of Proposition 2

We begin by mapping our notation into that of Jiang, Wu, Jin and Wong (2018) (JWJW, henceforth). Set $\sigma = 1$ without the loss of generality. Define the parameter vector θ by stacking the model parameters in $(\mathbf{W}, \mathbf{b})$, and define the function $\phi(\mathbf{x})$ by stacking $\mathbf{x}$ and the interactions of the variables in $\mathbf{x}$:

$$\theta \equiv \begin{pmatrix} \mathbf{b} \\ \mathbf{w}_{1\bullet} \\ \mathbf{w}_{2\bullet} \\ \vdots \\ \mathbf{w}_{n_1\bullet} \end{pmatrix} \in \mathbb{R}^{n+n_1n_2} \quad \text{and} \quad \phi(\mathbf{x}) \equiv \begin{pmatrix} \mathbf{x} \\ x_{11}\mathbf{x}_2 \\ x_{12}\mathbf{x}_2 \\ \vdots \\ x_{1n_1}\mathbf{x}_2 \end{pmatrix} \in \mathbb{R}^{n+n_1n_2}.$$

Let $\Theta \subset \mathbb{R}^d$ with $d \equiv n + n_1n_2$ denote the compact parameter space where θ is a priori known to lie. We distinguish the true model parameters, θ_* , by the asterisk.

Define the cumulant generating function $\Lambda(\theta) \equiv \log \sum_{\mathbf{x} \in \{0,1\}^n} e^{\theta^\top \phi(\mathbf{x})}$. The sBRE probability (3) of a participation profile $\mathbf{x}$ belongs to the **canonical exponential family** because it can be rewritten in the form $\mathbb{P}(\mathbf{x} \mid \theta) = e^{\theta^\top \phi(\mathbf{x}) - \Lambda(\theta)}$.

We now verify each of the assumptions (A1)–(A6) for JWJW’s Theorem 2.1 one by one.

Assumption (A1) requires that the parameter space Θ be a convex and compact subset of the natural parameter domain $\mathcal{D} \equiv \{\theta \in \mathbb{R}^d : \Lambda(\theta) < \infty\}$, and that the true parameter,

denoted here by θ_* , lie in the interior of Θ . The compactness of Θ and the interiority of θ_* are the maintained assumptions in our paper. The inclusion $\Theta \subset \mathcal{D}$ is immediate because $\Lambda(\theta) < \infty$ for all $\theta \in \mathbb{R}^d$.

Assumption (A2) requires that there exist a positive constant L such that, for all $\theta \in \Theta$, we have $\chi(\mathbb{P}(\cdot | \theta_*), \mathbb{P}(\cdot | \theta)) \leq L\|\theta_* - \theta\|$, where χ is a notion of divergence between two distributions described in JWW's Definition 2.1. JWW note that (A2) holds whenever Λ is in the exponential family with $\Lambda(\theta) < \infty$, which is the case in our model.

Informally, assumption (A3) requires that the best-response dynamics converge to the stationary distribution uniformly fast in θ . Formally, JWW state assumption (A3) as the requirement that a certain $\mathcal{L}^2$ -spectral gap be bounded away from zero uniformly in θ . In order to restate their condition for our finite setting, we introduce additional notation. Let $\mathbb{T}_\theta = (\mathbb{T}_\theta(\mathbf{x}, \mathbf{z}))_{\mathbf{x}, \mathbf{z}}$ denote the (ergodic) transition matrix of the Markov chain induced by the best-response dynamics in the two-sided market parameterized by θ . Let $\mathbf{h} = (h(\mathbf{x}))_{\mathbf{x}} \in \mathbb{R}^{2^n}$ denote a vector whose entries are indexed by a participation profile $\mathbf{x}$. Let $\mathbb{T}_\theta \mathbf{h} \in \mathbb{R}^{2^n}$ denote the vector defined by $(\mathbb{T}_\theta \mathbf{h})(\mathbf{x}) \equiv \sum_{\mathbf{z}} \mathbb{T}_\theta(\mathbf{x}, \mathbf{z}) h(\mathbf{z})$. Let $\langle \cdot, \cdot \rangle_\theta$ denote the inner product defined by $\langle \mathbf{h}, \mathbf{h}' \rangle_\theta \equiv \sum_{\mathbf{x} \in \{0,1\}^n} h(\mathbf{x}) h'(\mathbf{x}) \mathbb{P}(\mathbf{x} | \theta)$ for all vectors $\mathbf{h}$ and $\mathbf{h}'$. Let $\|\cdot\|_\theta$ denote the norm induced by $\langle \cdot, \cdot \rangle_\theta$. Define $\mathbf{1} = (1, \dots, 1)$. With this notation, JWW's assumption (A3) becomes

$$\sup_{\theta \in \Theta} \sup_{\|\mathbf{h}\|_\theta = 1} \{\|\mathbb{T}_\theta \mathbf{h}\|_\theta : \langle \mathbf{h}, \mathbf{1} \rangle_\theta = 0\} < 1, \quad (\text{A.6})$$

which—as we explain below—is a fancy way of specifying that the second largest eigenvalue modulus of the transition matrix $\mathbb{T}_\theta$ must be bounded away from 1 uniformly in θ . That is, the spectral gap must be positive. Because, for all the action revisions protocols that we consider, the Markov chain induced by the associated best-response dynamics is irreducible and aperiodic for all θ , the Perron–Frobenius theorem implies uniqueness of the largest eigenvalue, 1. This uniqueness immediately implies a positive spectral gap. This gap is bounded from zero uniformly in θ because θ is assumed to come from a compact set Θ , and because all eigenvalues are continuous in θ .

It remains to explain why the inequality in (A.6) is concerned with a spectral gap. Because the stationary distribution $\mathbb{P}(\cdot | \theta)$ satisfies the detailed balancedness condition (A.1), introduced and verified in the proof of Proposition 1, the operator $\mathbb{T}_\theta$ is self-adjoint (see, e.g., [Bremaud, 2020](#), Theorem 9.1.2, p. 290); that is, $\langle \mathbf{h}, \mathbb{T}_\theta \mathbf{h}' \rangle_\theta = \langle \mathbb{T}_\theta \mathbf{h}, \mathbf{h}' \rangle_\theta$ for all $\mathbf{h}$ and $\mathbf{h}'$. Then, by the spectral theorem, there exists an orthonormal basis of the space on which $\mathbb{T}_\theta$ acts: $\mathbb{T}_\theta = \sum_{i=1}^{2^n} \lambda_i \mathbf{v}_i \mathbf{v}_i^\top$, where each λ_i is an eigenvalue, and $\mathbf{v}_i$ is the corresponding eigenvector with $\|\mathbf{v}_i\|_\theta = 1$. (Even though the eigenvalues and the

eigenvectors depend on θ , we suppress this dependence for brevity.) Let $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_{2^n}|$ without the loss of generality. By the Perron–Frobenius theorem, the largest eigenvalue is $|\lambda_1| = 1$, with the corresponding eigenvector being $\mathbf{v}_1 = \mathbf{1}$, and the second-largest eigenvalue modulus is $|\lambda_2| < 1$. Therefore, for any $\mathbf{h}$ with $\langle \mathbf{h}, \mathbf{1} \rangle_\theta = 0$, there exist coefficients $\{c_i\}_{i=2}^{2^n}$ such that $\mathbf{h} = \sum_{i=2}^{2^n} c_i \mathbf{v}_i$. If, in addition, $\mathbf{h}$ satisfies $\|\mathbf{h}\|_\theta = 1$, then

$$\|\mathbb{T}_\theta \mathbf{h}\|_\theta = \left(\sum_{i=2}^{2^n} c_i^2 \lambda_i^2 \right)^{\frac{1}{2}} \leq |\lambda_2| \left(\sum_{i=2}^{2^n} c_i^2 \right)^{\frac{1}{2}} = |\lambda_2| \|\mathbf{h}\|_\theta = |\lambda_2|,$$

with equality if and only if $\mathbf{h} = \mathbf{v}_2$. Therefore, the inner sup in (A.6) equals the second largest eigenvalue modulus.

Assumption (A4) requires the random variable $\phi(\mathbf{x})$ when $\mathbf{x}$ is distributed according to a certain probability distribution on $\{0, 1\}^n$ to be subexponential (i.e., to abhor “fat tails”). Because each $\mathbf{x}$ in $\{0, 1\}^n$ is bounded, $\phi(\mathbf{x})$ is also bounded and, therefore, subexponential for all distributions of $\mathbf{x}$.

Assumption (A5) requires the expected potential after m steps of the best-response dynamics to depend continuously on θ . Formally, letting $\mathbf{x}^m$ denote the random variable that is the participation profile after m steps of the best-response dynamics, (A5) requires the expectation $\mathbb{E}[\theta^\top \phi(\mathbf{x}^m) \mid \mathbf{x}, \theta]$ to be Lipschitz continuous in θ for all $\mathbf{x}$. Note that, in our case, the m -step transition matrix is continuously differentiable in θ , and, therefore, the expectation is also continuous. A continuously differentiable function is Lipschitz continuous on the compact set Θ .

Assumption (A6) bounds the variance-covariance matrix of $\phi(\mathbf{x}^m)$ conditional on $\mathbf{x}$ and θ . Because all $\mathbf{x}$ in $\{0, 1\}^n$ are bounded, $\phi(\mathbf{x})$ is also bounded. The variance-covariance matrix is therefore also appropriately bounded.

A.3 Proof of Proposition 3

Consider an instance of the two-sided market in which each entry of the matrix $\mathbf{W}$ is 1, and each entry of the vector $\mathbf{b}$ is $-\beta$ for some positive scalar β that satisfies

$$\frac{1}{\beta} < \frac{1}{n_1} + \frac{1}{n_2} < \frac{2}{\beta}. \quad (\text{A.7})$$

For any participation profile $\mathbf{x}$, the potential satisfies:

$$\begin{aligned}
\mathbf{b}^\top \mathbf{x} + \mathbf{x}^\top_1 \mathbf{W} \mathbf{x}_2 &= \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} x_{1i} x_{2j} - \beta \left(\sum_{i=1}^{n_1} x_{1i} + \sum_{j=1}^{n_2} x_{2j} \right) \\
&= \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \left(x_{1i} x_{2j} - \frac{\beta x_{1i}}{n_2} - \frac{\beta x_{2j}}{n_1} \right) \\
&\leq \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \max_{(x,y) \in \{0,1\}^2} \left(xy - \frac{\beta x}{n_2} - \frac{\beta y}{n_1} \right) \\
&= n_1 n_2 \max \left\{ 0, 1 - \frac{\beta}{n_2} - \frac{\beta}{n_1} \right\} \\
&= 0,
\end{aligned}$$

where the last equality follows from the first inequality in (A.7). As a result, the potential is bounded above by zero. This upper bound is uniquely attained at $\mathbf{x} = \mathbf{0}$. Therefore, as $\sigma \rightarrow 0$, sBRE converges in probability to $\mathbf{x} = \mathbf{0}$, the maximizer of the potential. The associated welfare converges to zero.

For any participation profile $\mathbf{x}$, the utilitarian welfare satisfies:

$$\begin{aligned}
\mathbf{b}^\top \mathbf{x} + 2\mathbf{x}^\top_1 \mathbf{W} \mathbf{x}_2 &= 2 \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} x_{1i} x_{2j} - \beta \left(\sum_{i=1}^{n_1} x_{1i} + \sum_{j=1}^{n_2} x_{2j} \right) \\
&= \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \left(2x_{1i} x_{2j} - \frac{\beta x_{1i}}{n_2} - \frac{\beta x_{2j}}{n_1} \right) \\
&\leq n_1 n_2 \max_{x,y} \left(2xy - \frac{\beta x}{n_2} - \frac{\beta y}{n_1} \right) \\
&= n_1 n_2 \max \left\{ 0, 2 - \frac{\beta}{n_2} - \frac{\beta}{n_1} \right\} \\
&= n_1 n_2 \left(2 - \frac{\beta}{n_2} - \frac{\beta}{n_1} \right),
\end{aligned}$$

where the last equality follows from the second inequality in (A.7). As a result, the welfare is bounded above by the positive scalar $n_1 n_2 \left(2 - \frac{\beta}{n_2} - \frac{\beta}{n_1} \right)$. This upper bound is achieved at the participation profile $\mathbf{x} = (1, \dots, 1)$.

Conclude that, as $\sigma \rightarrow 0$, for the specified instance $(\mathbf{W}, \mathbf{b})$, the welfare at sBRE converges to zero, whereas the welfare at the utilitarian solution converges to a positive number. The price of anarchy—the ratio of the two limits—is thus infinite, as claimed.

A.4 Proof of Proposition 5

The proof is by example. Let $n_1 = n_2 = 2$, $\mathbf{b} = 0$, and

$$\mathbf{W} = \begin{pmatrix} w & -w \\ -w & w \end{pmatrix}, \quad \text{where } w > 0.$$

For brevity, we shall refer to side-1 agents 1 and 2 as agents 11 and 12, respectively, and to side-2 agents 1 and 2 as agents 21 and 22, respectively.

The sBRE is

$$\mathbb{P}(\mathbf{x} \mid \mathbf{W}, \mathbf{b}) = \frac{\exp w (x_{11} - x_{12}) (x_{21} - x_{22})}{12 + 2 \exp(w) + 2 \exp(-w)},$$

which implies that each agent participates with probability $\frac{1}{2}$, and that participation decisions are correlated, positively for agents 11 and 21, and 12 and 22, and negatively for agents 11 and 22, and 12 and 21. The platform identifies model parameters off this equilibrium behavior.

When the platform uses the correct model of the two-sided market—the one with heterogeneous network effects, the one that generates the data—it correctly identifies $(\mathbf{W}, \mathbf{b})$. The platform then computes a uniform access price that maximizes the expected revenue. This price, denoted $p^{true}(w)$, satisfies

$$p^{true}(w) \in \arg \max_p \Pi^{true}(p, w),$$

where

$$\Pi^{true}(p, w) \equiv \sum_{\mathbf{x} \in \{0,1\}^n} \mathbb{P}(\mathbf{x} \mid \mathbf{W}, (-p, -p, -p, -p)) (x_{11} + x_{12} + x_{21} + x_{22}) p.$$

The price $p^{true}(w)$ admits no explicit expression.

Now consider the naive platform that falsely believes that network effects are the same for all agents and equal u , so that the matrix of network effects is

$$\mathbf{U} = \begin{pmatrix} u & u \\ u & u \end{pmatrix}.$$

The naive platform believes the vector of standalone values to be some vector $\mathbf{a}$. As a result, the platform believes that, at sBRE, a participation profile $\mathbf{x}$ has the probability

$$\mathbb{P}^{miss}(\mathbf{x} \mid \mathbf{a}, \mathbf{U}) \equiv \frac{\exp(\mathbf{a}^\top \mathbf{x} + u(x_{11} + x_{12})(x_{21} + x_{22}))}{\sum_{\mathbf{y} \in \{0,1\}^n} \exp(\mathbf{a}^\top \mathbf{y} + u(y_{11} + y_{12})(y_{21} + y_{22}))}.$$

To identify the parameters of the misspecified model, the platform works out the likelihood function under this misspecified model and derives the counterparts to the first-order conditions in (4):

$$\begin{aligned} \sum_{\mathbf{x} \in \{0,1\}^n} \left(\mathbb{P}(\mathbf{x} \mid \mathbf{W}, \mathbf{b}) - \mathbb{P}^{miss}(\mathbf{x} \mid \mathbf{a}, \mathbf{U}) \right) (x_{11} + x_{12})(x_{21} + x_{22}) &= 0 \\ \sum_{\mathbf{x} \in \{0,1\}^n} \left(\mathbb{P}(\mathbf{x} \mid \mathbf{W}, \mathbf{b}) - \mathbb{P}^{miss}(\mathbf{x} \mid \mathbf{a}, \mathbf{U}) \right) x_{1i} &= 0, \quad i = 1, 2 \\ \sum_{\mathbf{x} \in \{0,1\}^n} \left(\mathbb{P}(\mathbf{x} \mid \mathbf{W}, \mathbf{b}) - \mathbb{P}^{miss}(\mathbf{x} \mid \mathbf{a}, \mathbf{U}) \right) x_{2j} &= 0, \quad j = 1, 2. \end{aligned}$$

The first equation in the display above directs the platform to the conclusion that the network effect u is greater whenever the correlation between the aggregate turnouts on the two sides of the market (i.e., the correlation between $x_{11} + x_{12}$ and $x_{21} + x_{22}$) is greater. In our example, however, turnouts are uncorrelated because agent 11's positive correlation with agent 21 is exactly offset by agent 11's negative correlation with agent 22, and agent 12's positive correlation with agent 22 is exactly offset by agent 12's negative correlation with agent 21. The naive platform thus (wrongly) concludes that no externality exists (i.e., that $\mathbf{U} = \mathbf{0}$). The naive platform (correctly) concludes from the equations in the display above that all standalone values from participation are zero (i.e., that $\mathbf{a} = \mathbf{0}$).

Having failed to detect any network effects whatsoever, all that the naive platform can hope to exploit through access pricing are the mistakes that the agents make when best responding. The platform optimally chooses an anonymous participation price p^{miss} that maximizes the expected revenue:

$$p^{miss} \in \arg \max_p \Pi^{miss}(p),$$

where

$$\Pi^{miss}(p) \equiv \sum_{\mathbf{x}} \mathbb{P}(\mathbf{x} \mid (-p, -p, -p, -p), \mathbf{0}) (px_{11} + px_{12} + px_{21} + px_{22}).$$

The price p^{miss} admits no explicit expression.

We now show that, as w grows, the actual revenue under p^{miss} as a fraction of the actual revenue under $p^{true}(w)$ converges to zero. That is, we show that

$$\lim_{w \rightarrow \infty} \frac{\Pi^{true}(p^{miss})}{\Pi^{true}(p^{true}(w))} = 0.$$

The argument proceeds in a sequence of steps that exploit the properties of the functions Π^{miss} and Π^{true} ; these properties, stated below without proof, can all be verified analytically:

- $\lim_{p \rightarrow \infty} \Pi^{miss}(p) = 0$ and $\Pi^{miss}(p) > 0$ for some p imply $p^{miss} < \infty$;
- $\lim_{w \rightarrow \infty} \Pi^{true}(p, w) = 2p$ for all p and $p^{miss} < \infty$ imply $\lim_{w \rightarrow \infty} \Pi^{true}(p^{miss}, w) < \infty$;
- $\lim_{w \rightarrow \infty} \Pi^{true}(w/3, w) = \infty$ implies $\lim_{w \rightarrow \infty} \Pi^{true}(p^{true}(w), w) = \infty$;
- $\lim_{w \rightarrow \infty} \Pi^{true}(p^{miss}, w) < \infty$ and $\lim_{w \rightarrow \infty} \Pi^{true}(p^{true}(w), w) = \infty$ imply

$$\lim_{w \rightarrow \infty} \Pi^{true}(p^{miss}) / \Pi^{true}(p^{true}(w)) = 0,$$

which proves the proposition.

A.5 Proof of Remark 1

First, we extend the model by assuming that agents on side 1 of the market capture a fraction θ_1 of the bilateral benefit from pairwise interaction, whereas agents on side 2 capture a fraction θ_2 , with $\theta_1 + \theta_2 = 1$. This is how asymmetry is usually modeled in the two-sided markets literature. The burden of this extension is purely notational; no conceptual innovation is required to extend our results to this more general case.

The total benefit from the bilateral interaction between agents i and j on sides 1 and 2 of the market is $2W_{ij}$, just as before. Now, however, i and j need not share this benefit equally. Instead, agent i , on side 1, collects $2\theta_1 W_{ij}$ in interaction benefits, and his counterparty j collects $2\theta_2 W_{ij}$, the remainder.

The standalone values from participation for agent i on side 1 and agent j on side 2 are, respectively, $2\theta_1 b_{1i}$ and $2\theta_2 b_{2j}$, where $2\theta_1$ and $2\theta_2$ are the scaling factors. With this new notation—but without prices—agent i 's and agent j 's payoffs are

$$2\theta_1 x_{1i} \left(b_{1i} + \mathbf{x}_2^\top \mathbf{W}_{i\bullet} \right) \quad \text{and} \quad 2\theta_2 x_{2j} \left(b_{2j} + \mathbf{x}_1^\top \mathbf{W}_{\bullet j} \right).$$

Agent i 's payoff is strategically equivalent to $x_{1i} (b_{1i} + \mathbf{x}_2^\top \mathbf{W}_{i\bullet})$, which is the same as in the baseline model. Ditto for agent j . As a result, CD consistently estimates $(\mathbf{W}, \mathbf{b})$, just as it did before. The underlying payoff parameters are then readily recovered from $(\mathbf{W}, \mathbf{b})$ by performing the requisite multiplication by $2\theta_1$ and $2\theta_2$.

Prices are now assumed to be parameterized by price-parameters $(\mathbf{P}, \mathbf{p})$. Agent i on side 1 pays $2\theta_1 p_{1i}$ for access and $2\theta_1 P_{ij}$ for interacting with every agent j who shows up on side 2. Under a pricing scheme $(\mathbf{P}, \mathbf{p})$, agent i 's payoff is

$$2\theta_1 x_{1i} \left(b_{1i} - p_{1i} + \mathbf{x}_2^\top (\mathbf{W}_{i\bullet} - \mathbf{P}_{i\bullet}) \right).$$

This payoff is strategically equivalent to $x_{1i} (b_{1i} - p_{1i} + \mathbf{x}_2^\top (\mathbf{W}_{i\bullet} - \mathbf{P}_{i\bullet}))$, the same as in the baseline model. The situation of agent j is analogous. As a result, the probability of observing a participation profile $\mathbf{x}$ remains $\mathbb{P}(\mathbf{x} \mid \mathbf{W} - \mathbf{P}, \mathbf{b} - \mathbf{p})$, with $\mathbb{P}$ defined in (3). The revenue-maximization problem becomes

$$\max_{\mathbf{P}, \mathbf{p}} \sum_{\mathbf{x} \in \{0,1\}^n} \mathbb{P}(\mathbf{x} \mid \mathbf{W} - \mathbf{P}, \mathbf{b} - \mathbf{p}) \left(2\theta_1 \mathbf{p}_1^\top \mathbf{x}_1 + 2\theta_2 \mathbf{p}_2^\top \mathbf{x}_2 + 2\mathbf{x}_1^\top \mathbf{P} \mathbf{x}_2 \right).$$

Proposition A.1 restates Remark 1 formally.

Proposition A.1 (Do not Divide; Conquer). *In the model's extension described above, if the price-parameters have $\mathbf{P} = (1 - \sqrt{\sigma}) \mathbf{W}$, $\mathbf{p}_1 = (1 - \theta_1 \sqrt{\sigma}) \mathbf{b}_1$, and $\mathbf{p}_2 = (1 - \theta_2 \sqrt{\sigma}) \mathbf{b}_2$, then the equilibrium revenue converges to the first-best welfare as $\sigma \rightarrow 0$.*

Proof. One can verify that, under the price parameters specified in the proposition, the expected revenue approaches the first-best welfare $\max_{\mathbf{x}} \{2\theta_1 \mathbf{b}_1^\top \mathbf{x}_1 + 2\theta_2 \mathbf{b}_2^\top \mathbf{x}_2 + 2\mathbf{x}_1^\top \mathbf{W} \mathbf{x}_2\}$ as $\sigma \rightarrow 0$. ■

In Proposition A.1, agent i on side 1 pays $2\theta_1 (1 - \theta_1 \sqrt{\sigma}) b_{1i}$ for access to the market and approximately $2\theta_1 (1 - \sqrt{\sigma}) W_{ij}$ for interacting with every agent j who shows up on side 2. Recall that the corresponding benefits experienced by the agent are $2\theta_1 b_{1i}$ and $2\theta_1 W_{ij}$, respectively. As a result, for a “small” σ , each agent i parts with approximately his entire surplus. The platform conquers all and does not divide.

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